Policy Optimization: From CPI to PPO

Conservative Updates, Trust Regions, Advantage Estimation & Clipping

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Part I — Main Sources

The Policy Gradient Theorem

Sutton et al. (1999)

The foundation that proved you can optimize a parameterized policy directly by estimating the gradient of the expected return.

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s) Q^{\pi_\theta}(s,a)\right]$$



Neuro-Dynamic Programming

Bertsekas & Tsitsiklis (1996), Ch. 6

Formalized Approximate Policy Iteration (API). Chapter 6 is historically significant here because it exposed the fundamental instability of using function approximators (like neural networks) in reinforcement learning.

Conservative Policy Iteration

Kakade & Langford (2002)

Established the theoretical foundation for monotonic policy improvement using the performance difference lemma and mixture policies.

Trust Region Policy Optimization

Schulman et al. (2015)

Scaled CPI's concepts to deep neural networks by using a KL-divergence constraint to enforce a "trust region" during policy updates, preventing destructively large steps.

$$J(\pi_{new}) \geq L_{\pi_{old}}(\pi_{new}) - \frac{4\gamma\epsilon}{(1-\gamma)^2}\max_s D_{KL}(\pi_{old}(\cdot|s)\|\pi_{new}(\cdot|s))$$

Approximately Optimal Approximate Reinforcement Learning

Kakade & Langford
ICML 2002

Introduces **Conservative Policy Iteration (CPI)**

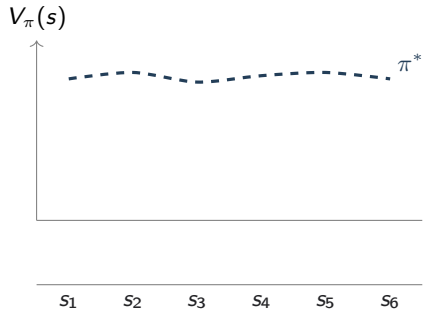
Setting & Assumptions

A finite MDP $(S, D, A, \mathcal{R}, P)$:

- **Finite** state and action spaces: $|S|, |A| < \infty$.
- **Bounded rewards**: $\mathcal{R} : S \times A \to [0, R]$.
- **Discounted**: $\gamma \in [0, 1)$.
- **Unknown dynamics**: $P(s'; s, a)$ is not given — we can only sample transitions (model-free).
- **Start-state distribution** D: the goal is

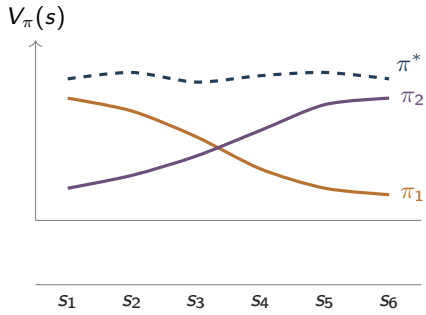
$$\eta_D(\pi) = \mathbb{E}_{s_0 \sim D}\left[V_\pi(s_0)\right]$$

The Start-State Distribution D



Exact: π^* on top everywhere
$\Rightarrow$ D irrelevant.

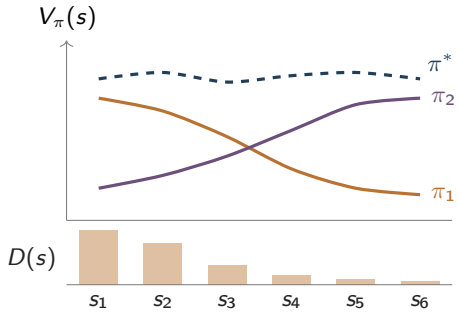
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Approximate: curves cross
restricted class $\{\pi_\theta\}$ · limited data

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Goal (Sec. 2): rank by D

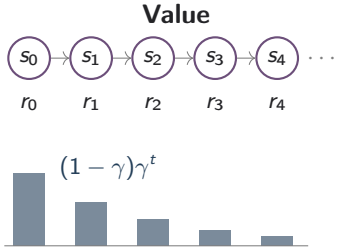
$$\eta_D(\pi) = \sum_s D(s)\, V_\pi(s)$$

Here π_1 wins.

D **comes from the application:** robot start poses, game openings, user prompts.

Notation

$$V_\pi(s) = (1-\gamma)\,\mathbb{E}\Big[\sum_t \gamma^t \mathcal{R}(s_t, a_t) \,\Big|\, \pi, s_0{=}s\Big]$$



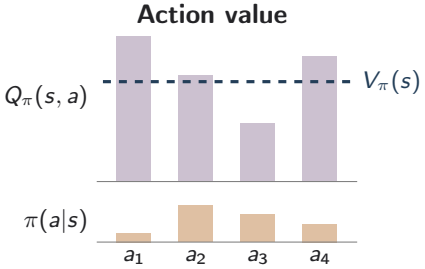
Weights sum to 1, so V is an average of rewards: $V \in [0, R]$.

Notation

$$V_\pi(s) = (1-\gamma)\,\mathbb{E}\Big[\sum_t \gamma^t \mathcal{R}(s_t, a_t) \,\Big|\, \pi, s_0{=}s\Big]$$

$$Q_\pi(s,a) = (1-\gamma)\mathcal{R}(s,a) + \gamma\,\mathbb{E}_{s'}\big[V_\pi(s')\big]$$

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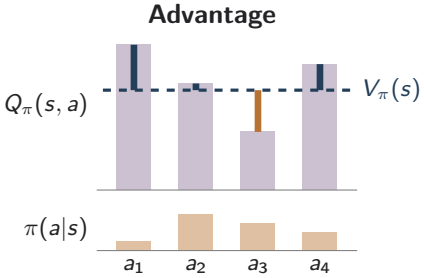
Take a first, then follow π.
V_π = bars averaged with weights $\pi(a|s)$.

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$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$$
$$\mathbb{E}_{a\sim\pi}\big[A_\pi(s,a)\big] = 0, \quad \mathbb{E}_{a\sim\pi'}\big[A_\pi(s,a)\big] \neq 0$$



A = gap to the line. Weighted by π the gaps cancel.
A π' that favors the **positive** gaps gets $\mathbb{E}_{\pi'}[A_\pi] > 0$.

Notation

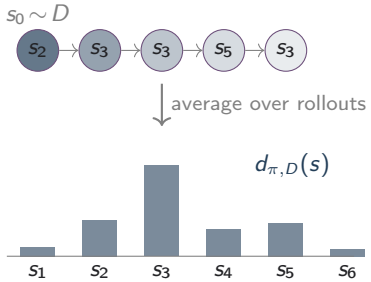
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$$d_{\pi,D}(s) = (1-\gamma)\sum_t \gamma^t \Pr(s_t = s;\, \pi, D)$$

Discounted visitation



Where π spends its time from $s_0 \sim D$, early steps weighted more.
Not the same as D (only s_0).

Notation

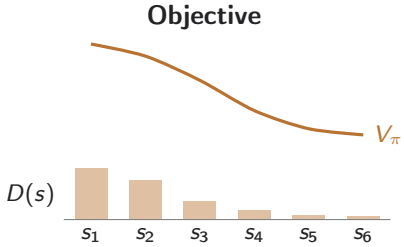
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$$d_{\pi,D}(s) = (1-\gamma)\sum_t \gamma^t \,\mathrm{Pr}(s_t = s;\, \pi, D)$$

$$\eta_D(\pi) = \mathbb{E}_{s\sim D}\big[V_\pi(s)\big]$$



Average V_π over start states only.
One number to rank policies.

Three Questions Any Approximate Method Should Answer

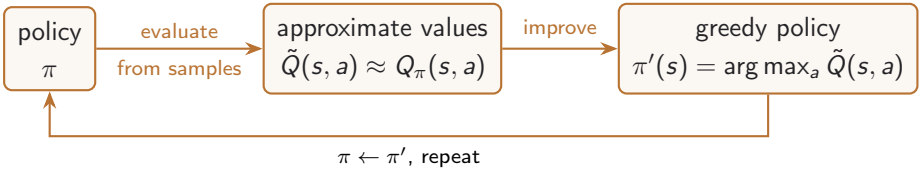
1. Is there a **performance measure** that is guaranteed to **improve at every step**?
2. How hard is it to **verify** that a given update improves this measure?
3. What **performance level** is reached after a **reasonable number** of updates?

Note: not "what is the asymptotic policy?" — only policies reachable in reasonable time.

Approximate value functions
Weak answers to all three.

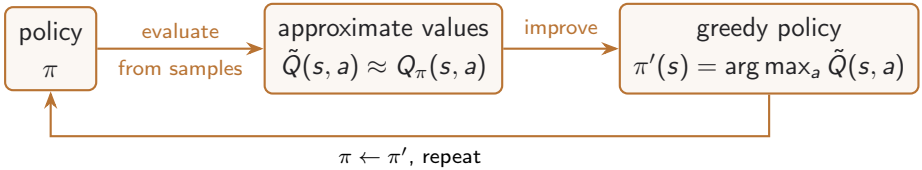
Policy gradients
Good answer to (1), poor answers to (2) and (3).

Reminder: Approximate Value Function Methods



Exact policy iteration, with Q_π replaced by an **estimate** $\tilde{Q}$.

Reminder: Approximate Value Function Methods



Exact policy iteration, with Q_π replaced by an **estimate** $\tilde{Q}$.

Example: Q-learning with a function approximator $\tilde{Q}_w$

$$w \leftarrow w + \alpha \left(y_t - \tilde{Q}_w(s_t, a_t)\right) \nabla_w \tilde{Q}_w(s_t, a_t)$$

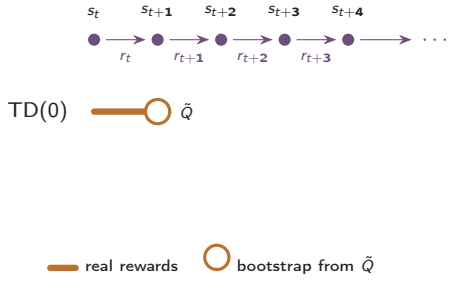
$$y_t = (1-\gamma)\, r_t + \gamma \max_{a'} \tilde{Q}_w(s_{t+1}, a')$$

The max is the greedy step, so evaluate and improve happen together.
The target y_t is one choice of many $\rightarrow$ next slide.

How Is $\tilde{Q}$ Estimated? From TD(0) to Monte Carlo

TD(0): one reward, then bootstrap

$$y_t^{(1)} = (1-\gamma)\, r_t + \gamma\, \tilde{Q}(s_{t+1}, a_{t+1})$$



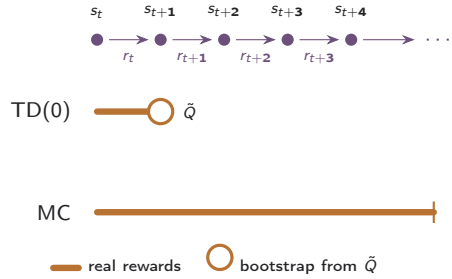
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Monte Carlo: the whole observed return

$$y_t^{(\infty)} = (1-\gamma) \sum_{k \geq 0} \gamma^k r_{t+k}$$



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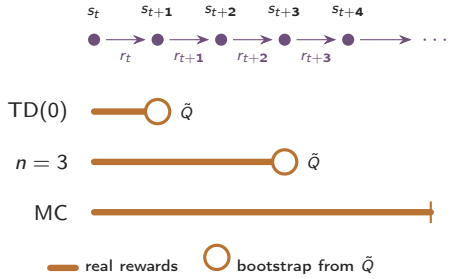
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n-step: in between

$$y_t^{(n)} = (1-\gamma) \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n\, \tilde{Q}(s_{t+n}, a_{t+n})$$



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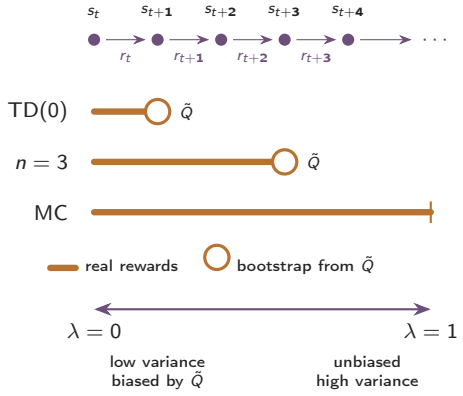
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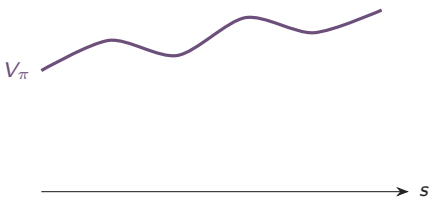
$$y_t^{(n)} = (1-\gamma) \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n\, \tilde{Q}(s_{t+n}, a_{t+n})$$

TD(λ): average over all n

$$y_t^{\lambda} = (1-\lambda) \sum_{n \geq 1} \lambda^{n-1}\, y_t^{(n)}$$



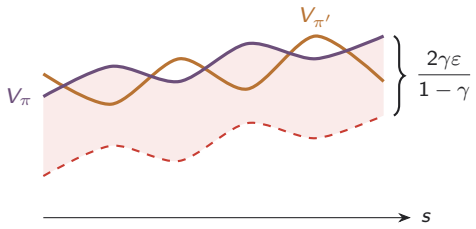
Gap 1: Approximate Value Functions Give No Improvement Guarantee



Approximation error, measured **everywhere** (ℓ_∞):

$$\varepsilon = \max_s \left| \tilde{V}(s) - V_\pi(s) \right|$$

Gap 1: Approximate Value Functions Give No Improvement Guarantee



$V_{\pi'}$ may sit anywhere above the dashed line — **also below** V_π.

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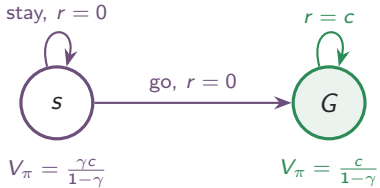
With π' greedy w.r.t. $\tilde{V}$ (Bertsekas & Tsitsiklis):

$$V_{\pi'}(s) \;\geq\; V_\pi(s) - \frac{2\gamma\varepsilon}{1-\gamma} \qquad \forall s$$

Only "won't get **much worse**" — never "gets better".

Gap 1: The Bound Is Tight

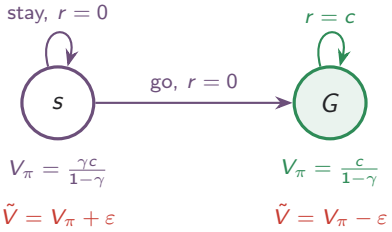
1. π goes to G and collects c forever.



π = go. Set $c = 2\varepsilon$.

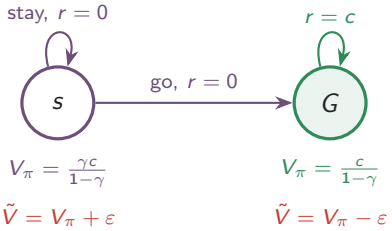
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1. π goes to G and collects c forever.
2. $\tilde{V}$ is off by ε: too high at s, too low at G.



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Gap 1: The Bound Is Tight



π = go. Set $c = 2\varepsilon$.

1. π goes to G and collects c forever.
2. $\tilde{V}$ is off by ε: too high at s, too low at G.
3. Greedy w.r.t. $\tilde{V}$ at s:

$$\underbrace{\gamma\big(V_\pi(s) + \varepsilon\big)}_{\text{stay}} = \underbrace{\gamma\big(V_\pi(G) - \varepsilon\big)}_{\text{go}}$$

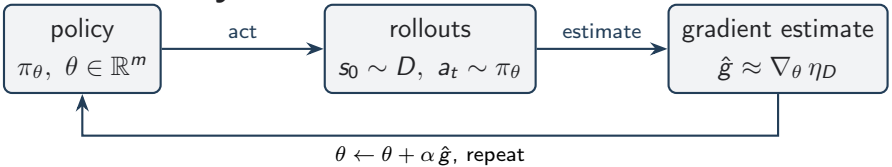
a tie, since $V_\pi(G) - V_\pi(s) = c = 2\varepsilon$, so π' may stay forever: $V_{\pi'}(s) = 0$.

At s: $V_\pi(s) - V_{\pi'}(s) = \dfrac{\gamma c}{1-\gamma} = \dfrac{2\gamma\varepsilon}{1-\gamma}$ — exactly the bound.

At G: one action only, so $\pi' = \pi$ and the loss is 0.

The bound is a max over states; s attains it.

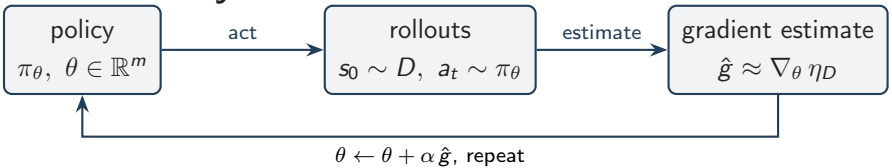
Reminder: Policy Gradient Methods



$$\nabla_\theta \eta_D = \frac{1}{1-\gamma} \sum_{s,a} d_{\pi,D}(s)\, \nabla_\theta \pi(a;s)\, Q_\pi(s,a)$$

Small steps uphill on η_D itself $\Rightarrow$ a good answer to question 1.

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Example: REINFORCE with a Gibbs (softmax) policy

$$\pi_\theta(a;s) = \frac{e^{\theta_{sa}}}{\sum_b e^{\theta_{sb}}}, \qquad \hat{g} \ \propto \ \sum_t \nabla_\theta \log \pi_\theta(a_t;s_t)\, \hat{Q}_t$$

$\hat{Q}_t$: return observed after step t. Same policy class as Example 2.

▸ Gibbs policy

Policy Gradient Is Stochastic Gradient Ascent

	Stochastic gradient descent		Policy gradient
objective	$\min_w \; J(w) = \mathbb{E}_X\big[f(w, X)\big]$	$\leftrightarrow$	$\max_\theta \; \eta_D(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\big[R(\tau)\big]$
looking for	$\nabla J(w) = \mathbb{E}\big[\nabla_w f(w, X)\big] = 0$	$\leftrightarrow$	$\nabla_\theta \, \eta_D(\theta) = 0$

Policy Gradient Is Stochastic Gradient Ascent

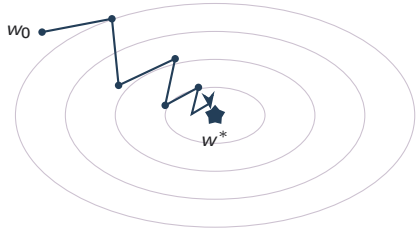
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one sample	$x_k \sim X$	$\leftrightarrow$	rollout τ_k: $s_0 \sim D, \;\; a_t \sim \pi_{\theta_k}$
update	$w_{k+1} = w_k - a_k \, \nabla_w f(w_k, x_k)$	$\leftrightarrow$	$\theta_{k+1} = \theta_k + a_k \, \hat{g}_k$

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noise	$\eta_k = \nabla_w f(w_k, x_k) - \nabla J(w_k)$	$\leftrightarrow$	$\eta_k = \hat{g}_k - \nabla_\theta \, \eta_D(\theta_k)$

What we know about SGD carries over. Twist: samples come from π_θ itself, yet the log-derivative trick (REINFORCE) keeps $\hat{g}_k$ **unbiased**. ▸ Why?

SGD Convergence: the Convex, Smooth Case



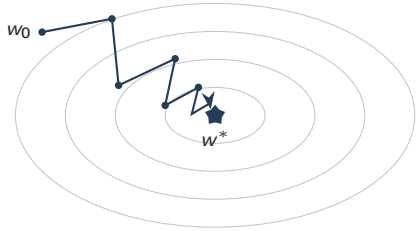
noisy steps, shrinking step size

Curvature: strongly convex and smooth

$$0 < c_1 \leq \nabla_w^2 f(w, X) \leq c_2$$

$\rightarrow$ one minimizer, and bounds on the rate

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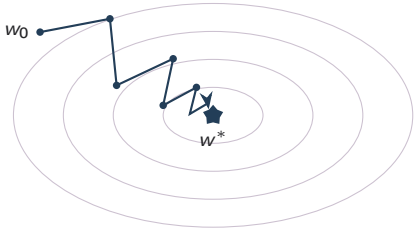
Unbiased noise

$$\mathbb{E}[\eta_k \mid \mathcal{H}_k] = 0$$

$\mathcal{H}_k$: the history so far, $w_0, x_0, \ldots, x_{k-1}$

$\rightarrow$ steps point the right way on average:
the estimate converges to w^*

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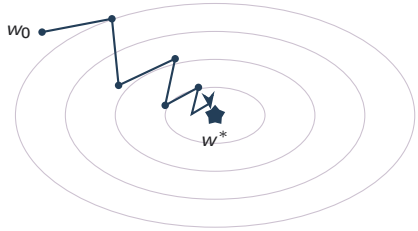
Finite variance and **step sizes**

$$\mathbb{E}[\eta_k^2 \mid \mathcal{H}_k] < \infty$$

$$\sum_k a_k = \infty, \quad \sum_k a_k^2 < \infty$$

$\rightarrow$ steps reach any distance, summed noise stays finite:
noise is filtered out relative to the signal

SGD Convergence: the Convex, Smooth Case



noisy steps, shrinking step size

$w_k \to w^*$ **almost surely**

Curvature: strongly convex and smooth

$$0 < c_1 \leq \nabla_w^2 f(w, X) \leq c_2$$

$\to$ one minimizer, and bounds on the rate

Unbiased noise

$$\mathbb{E}[\eta_k \mid \mathcal{H}_k] = 0$$

$\mathcal{H}_k$: the history so far, $w_0, x_0, \ldots, x_{k-1}$

$\to$ steps point the right way on average:
the estimate converges to w^*

Finite variance and **step sizes**

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Sample Complexity: What Theory Guarantees

$$N(\epsilon, \delta) = \min \left\{ N \in \mathbb{N} \;\middle|\; \mathbb{P}\big(\|w_k - w^*\| \leq \epsilon\big) \geq 1 - \delta \;\text{ for all } k \geq N \right\}$$

samples needed to be ϵ-close with probability $\geq 1 - \delta$ (or in value: $\mathbb{P}(J(w_k) - J(w^*) \leq \epsilon) \geq 1 - \delta$)

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policy gradients with neural nets →

Convex, smooth

global optimum: $N(\epsilon, \delta) = O\big(\sigma^2/(c_1^2 \epsilon^2 \delta)\big)$

Non-convex, smooth

stationary point only: $\mathbb{E}\|\nabla J(w_R)\| \leq \epsilon$ after $K = O(\sigma^2/\epsilon^4)$, w_R: a random iterate, $R \leq K$

Non-convex, non-smooth

weaker still: near-stationary points, special classes

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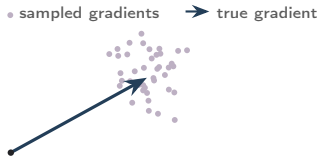
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Non-convex, non-smooth	weaker still: near-stationary points, special classes

Still useful: the convex case shows **what drives** N — the noise σ^2 and the curvature — and that carries over.

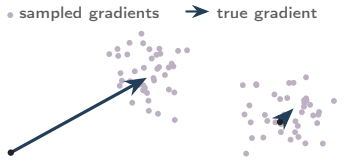
The Signal-to-Noise Reality



large $\|\nabla J\|$:
direction is clear

- Finite variance is enough for convergence *in the limit*.

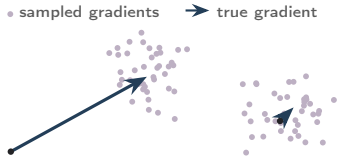
The Signal-to-Noise Reality



- Finite variance is enough for convergence *in the limit*.
- In practice N depends on the **signal-to-noise ratio**. Averaging n samples leaves noise $\sigma/\sqrt{n}$, so just for the *direction*:

$$n \gtrsim \sigma^2/\|\nabla J\|^2$$

The Signal-to-Noise Reality



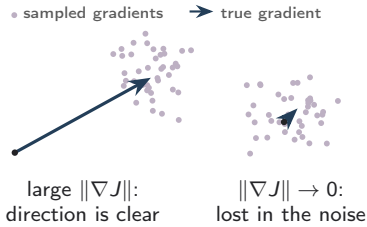
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- **Link to curvature**: strong convexity gives $\|\nabla J(w)\| \geq c_1\|w - w^*\|$, so at distance ϵ:

$$n \gtrsim \sigma^2/(c_1\epsilon)^2 \quad = \text{the convex-row } N \text{ (without } 1/\delta)$$

The Signal-to-Noise Reality



Next: two MDPs with an (almost) zero gradient.

- Finite variance is enough for convergence *in the limit*.
- In practice N depends on the **signal-to-noise ratio**. Averaging n samples leaves noise $\sigma/\sqrt{n}$, so just for the *direction*:

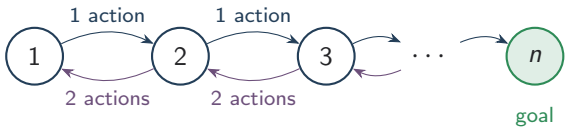
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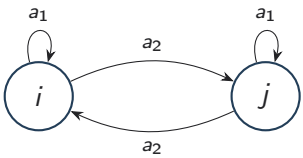
- Flat (c_1 small) or $\|\mathbb{E}[\nabla_w f(w, X)]\| \to 0$: noise dominates, **intractable** n.

Gap 2: Policy Gradients — Example 1 (Exploration)



- Uniform random policy: expected time to goal $= 3(2^n - n - 1) \approx 10^{15}$ for $n = 50$.
- Before reaching the goal, every sensible gradient estimate is 0: right *magnitude*, but **no direction**.
- Importance sampling doesn't help: weights would be exponentially large.

Gap 2: Policy Gradients — Example 2 (Plateaus)

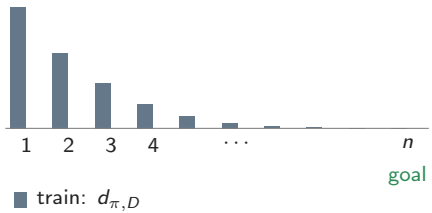


Gibbs policy: $\pi(a; s) \propto \exp(\theta_{sa})$

- Start: $\rho(i) = 0.8,\ \rho(j) = 0.2$.
- Raising the self-loop at i **starves** state j.
- $\rho(j)$ drops to $\approx 10^{-7} \Rightarrow$ nothing is learned at j.
- Result: an extremely **flat plateau** (on a 10^7 time scale).

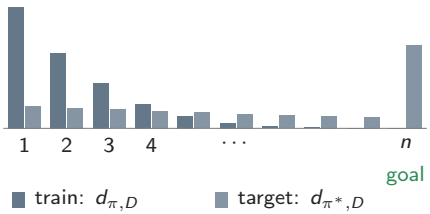
Root cause: η_D weighs improvements by $d_{\pi,D}$ — it is **insensitive to improvement at unlikely states**, even when those states are needed for near-optimal performance.

Gap 2 as Covariate Shift



Train: samples come from where the **current** π goes.

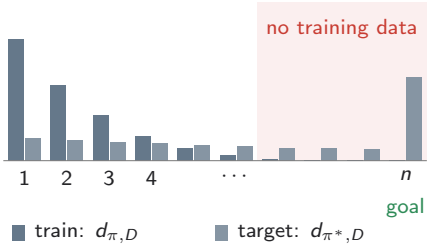
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Gap 2 as Covariate Shift

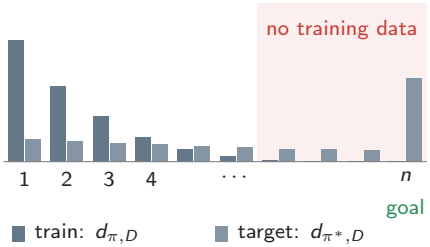


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Gap 2 as Covariate Shift



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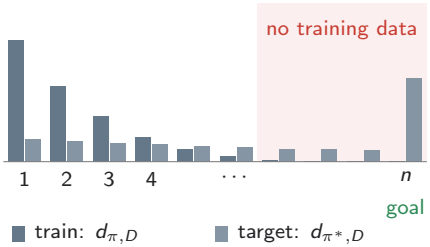
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Unlike standard covariate shift:

- target $d_{\pi^*,D}$ needs π^* — **unknown**
- data come from π itself — the gap **grows** as π learns (Ex. 2)

Gap 2 as Covariate Shift



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Unlike standard covariate shift:

- target $d_{\pi^*,D}$ needs π^* — **unknown**
- data come from π itself — the gap **grows** as π learns (Ex. 2)

Instead: change *where the data comes from*: train from μ, still judged by η_D.

$\rightarrow$ restart distribution μ (next slide)

The Two Fixes

1. Conservative update

Don't jump to the greedy policy — mix:

$$\pi_{\text{new}}(a; s) = (1 - \alpha)\,\pi(a; s) + \alpha\,\pi'(a; s)$$

$\alpha \in [0, 1]$ acts as a **step size**. $\alpha = 1$ is greedy policy iteration.

$\rightarrow$ addresses Gap 1 (greedy jumps)

2. Restart distribution μ

Optimize a measure that weighs states more uniformly:

$$\eta_\mu(\pi) = \mathbb{E}_{s\sim\mu}\big[V_\pi(s)\big]$$

μ is "exploratory"; the agent can restart from $s \sim \mu$ (weaker than a generative model*).

D: deployment (given) · μ: training resets (chosen)

$\rightarrow$ addresses Gap 2 (exploration)

- Any optimal policy maximizes both η_μ and η_D — but within a restricted class they can differ.

*Generative model (Kearns et al., 1999): sample $s' \sim P(\cdot \mid s, a)$ at *any* (s, a) you choose. A μ-restart only resets to a *random* $s \sim \mu$. Access, strongest to weakest: full P > generative model > μ-restart > one trajectory, no resets.

Roadmap: From the Two Fixes to the Three Answers

the two fixes

1. Conservative update
mix with a step α

2. Restart distribution
measure η_μ,
restarts $s \sim \mu$

results (next slides)

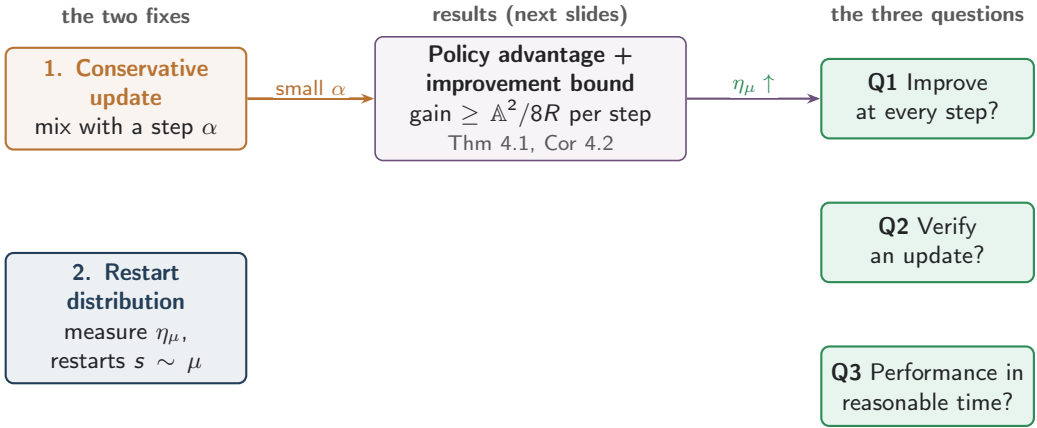
the three questions

Q1 Improve at every step?

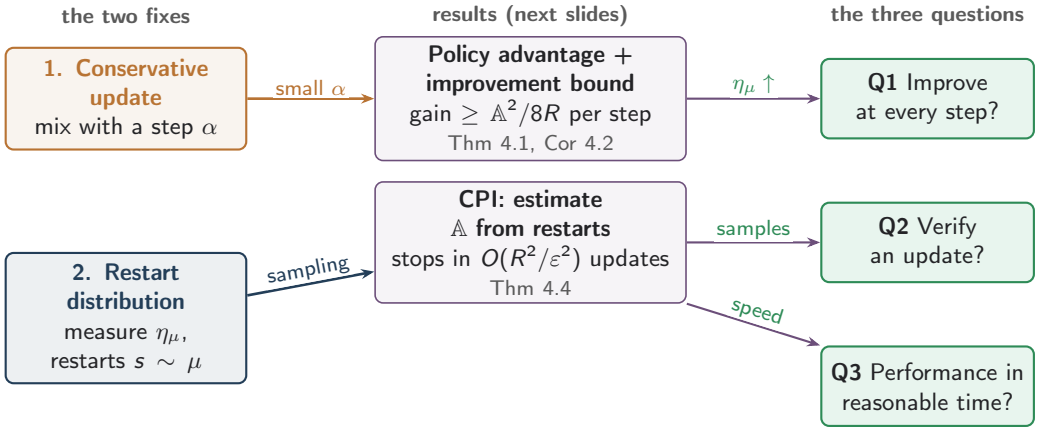
Q2 Verify an update?

Q3 Performance in reasonable time?

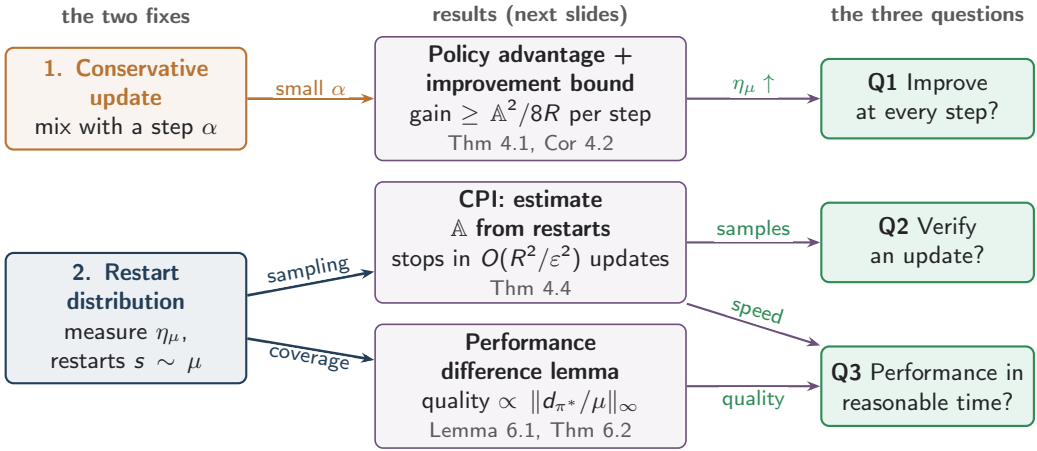
Roadmap: From the Two Fixes to the Three Answers



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Roadmap: From the Two Fixes to the Three Answers



Warm and cool arrows: where each fix enters. μ sets the **quality**, not the **speed**.

The Performance Difference Lemma

Per-state advantage of a policy π' over π — one step of π' at s, then back to π:

$$\bar{A}(s) \;=\; \mathbb{E}_{a\sim\pi'(\cdot;s)}\big[A_\pi(s,a)\big]$$

Lemma 6.1. For any π, π' and start distribution μ:

▸ Second equality

$$\eta_\mu(\pi') - \eta_\mu(\pi) \;=\; \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim d_{\pi',\mu}}\big[\bar{A}(s)\big] \;=\; \mathbb{E}\Big[\textstyle\sum_t \gamma^t\,\bar{A}(s_t)\;\Big|\; s_0\sim\mu,\ \pi'\Big]$$

The Performance Difference Lemma

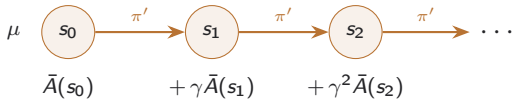
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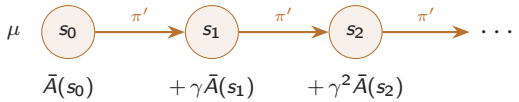
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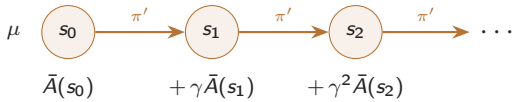
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- **Exact** — the gain is the one-step advantages, collected along the **new** policy's path.
- **Not measurable:** it needs $d_{\pi',\mu}$, the states of a policy we have not run yet.

A New Quantity: the Policy Advantage

$$\mathbb{A}_{\pi,\mu}(\pi') \;=\; \mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\mathbb{E}_{a\sim\pi'(\cdot;s)}\big[A_\pi(s,a)\big]\Big] \;=\; \mathbb{E}_{s\sim d_{\pi,\mu}}\big[\bar{A}(s)\big]$$

- Lemma 6.1's average, but **on the states π visits** (from μ) — so it is **measurable**.
- One step of policy improvement maximizes it.
- It is exactly the **first-order** change of η_μ in α:

▸ Proof

$$\left.\frac{\partial \eta_\mu}{\partial \alpha}\right|_{\alpha=0} = \frac{1}{1-\gamma}\mathbb{A}_{\pi,\mu}(\pi') \qquad \Longrightarrow \qquad \Delta\eta_\mu = \frac{\alpha}{1-\gamma}\mathbb{A}_{\pi,\mu}(\pi') + O(\alpha^2)$$

Small α + positive policy advantage $\Rightarrow$ improvement. Large $\alpha \Rightarrow$ possible degradation. How large can α be?

The Exact Change vs. the Policy Advantage

Apply Lemma 6.1 to the mixture $\pi_{\text{new}} = (1-\alpha)\pi + \alpha\pi'$: per state, $\mathbb{E}_{a\sim\pi_{\text{new}}} A_\pi(s,a) = \alpha\,\bar{A}(s)$.

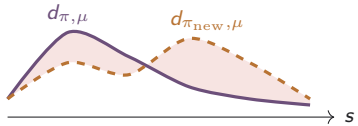
$$(1-\gamma)\big(\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi)\big) \;=\; \alpha \sum_s d_{\pi_{\text{new}},\mu}(s)\,\bar{A}(s)$$

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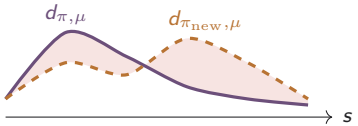
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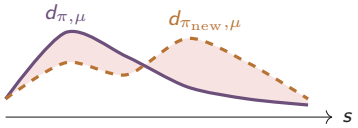
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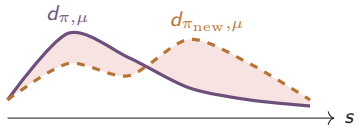
- $\alpha\mathbb{A}$: the first-order term from the policy-advantage slide.
- Gap: two factors of α — smaller $\bar{A}$ **and** less shift $\Rightarrow O(\alpha^2)$.

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shaded: the shift — $\bar{A}$ unknown there.

- $\alpha\mathbb{A}$: the first-order term from the policy-advantage slide.
- Gap: two factors of α — smaller $\bar{A}$ **and** less shift $\Rightarrow O(\alpha^2)$.
- Thm 4.1 = a **lower bound on the gap**.

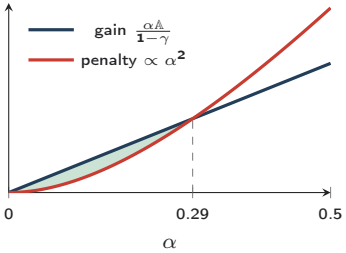
Exact but unmeasurable (Lemma 6.1) = measurable ($\alpha\mathbb{A}$) + gap. We only need to bound the gap.

How Big a Step? The Improvement Bound

Theorem 4.1. With $\varepsilon = \max_s \left|\mathbb{E}_{a\sim\pi'}[A_\pi(s,a)]\right|$, for all $\alpha \in [0,1]$:

$$\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \;\geq\; \frac{\alpha}{1-\gamma}\left(\mathbb{A} - \frac{2\alpha\gamma\varepsilon}{1-\gamma(1-\alpha)}\right)$$

- Gain $\alpha\mathbb{A}$: **linear**, measured on old states.
- Penalty: worst case of the state shift, $\propto \alpha^2$.
- Small α: the gain wins $\Rightarrow$ **guaranteed progress**.
- $\alpha = 1$: B&T penalty **plus a gain**. Compare



Illustrative: $\gamma = 0.6$, $\varepsilon = 1$, $\mathbb{A} = 0.6$. Green: improvement guaranteed.

Some small step always helps when $\mathbb{A} > 0$. **Which** α? $\rightarrow$ next slide.

Choosing the Step: Corollary 4.2

1. **Simplify:** lower the bound further, using $|A_\pi| \le R$, so that only the measurable $\mathbb{A}$ is left.

$$\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \;\ge\; \frac{\alpha}{1-\gamma}\Big(\mathbb{A} - \frac{2\alpha\gamma\varepsilon}{1-\gamma(1-\alpha)}\Big) \qquad \text{Thm 4.1}$$

$$\;\ge\; \frac{\alpha}{1-\gamma}\Big(\mathbb{A} - \frac{2\alpha R}{1-\gamma}\Big) \;=:\; f(\alpha) \qquad \varepsilon \le R,\;\; 1-\gamma(1-\alpha) \ge 1-\gamma,\;\; \gamma \le 1$$

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2. Maximize: $f(\alpha) = \frac{\mathbb{A}}{1-\gamma}\,\alpha - \frac{2R}{(1-\gamma)^2}\,\alpha^2$ is a downward parabola in α.

$$f'(\alpha) \;=\; \frac{\mathbb{A}}{1-\gamma} - \frac{4R\,\alpha}{(1-\gamma)^2} \;=\; 0 \quad\Longrightarrow\quad \alpha^* \;=\; \frac{(1-\gamma)\mathbb{A}}{4R}$$

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$$f(\alpha^*) \;=\; \frac{\mathbb{A}}{1-\gamma}\cdot\frac{(1-\gamma)\mathbb{A}}{4R} \;-\; \frac{2R}{(1-\gamma)^2}\cdot\frac{(1-\gamma)^2\mathbb{A}^2}{16R^2} \;=\; \frac{\mathbb{A}^2}{8R}$$

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Corollary 4.2. If $\mathbb{A} \ge 0$, then $\alpha = \frac{(1-\gamma)\mathbb{A}}{4R}$ guarantees $\eta_\mu(\pi_{\text{new}}) - \eta_\mu(\pi) \ge \frac{\mathbb{A}^2}{8R}$.

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Computable: $\mathbb{A}$ averages over the *old* policy's states (μ-restarts); R is known. Price: $\mathbb{A} \le R$ gives $\alpha \le (1-\gamma)/4$ (≤ 0.0025 at $\gamma = 0.99$), and $\alpha \to 0$ as $\mathbb{A} \to 0$: **tiny steps**.

Conservative Policy Iteration

1. $\pi' \leftarrow G_\varepsilon(\pi, \mu)$ (greedy policy chooser)
2. Estimate $\hat{\mathbb{A}} \approx \mathbb{A}_{\pi,\mu}(\pi')$ to $\pm\frac{\varepsilon}{3}$ with $O\big(\frac{R^2}{\varepsilon^2} \log \frac{R^2}{\delta\varepsilon^2}\big)$ μ-restarts
3. If $\hat{\mathbb{A}} < \frac{2\varepsilon}{3}$: **stop**, return π
4. Else: mix with $\alpha = \frac{(1-\gamma)(\hat{\mathbb{A}} - \varepsilon/3)}{4R}$, go to 1

G_ε: returns π' with $\mathbb{A}_{\pi,\mu}(\pi') \geq \max_{\pi''} \mathbb{A}_{\pi,\mu}(\pi'') - \varepsilon$ — e.g. a regression on advantages with *average* error $\leq \varepsilon/2$.

▸ How?

Theorem 4.4. With prob. $\geq 1 - \delta$, CPI:

i) improves η_μ at **every** update;

ii) stops within $72\, R^2/\varepsilon^2$ calls to G_ε;

iii) returns π with $\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) < 2\varepsilon$.[1]

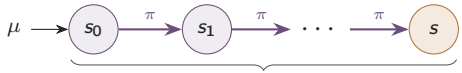
Number of updates does **not** depend on μ or on the state-space size.

[1] $\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) \equiv \max_{\pi'} \mathbb{A}_{\pi,\mu}(\pi') = \mathbb{E}_{s\sim d_{\pi,\mu}}\big[\max_a A_\pi(s,a)\big]$: the largest policy advantage any π' can achieve (Def. 4.3). Small $\Rightarrow$ no policy improves much on π under $d_{\pi,\mu}$.

Verifying an Update: Estimating $\mathbb{A}$ From μ-Restarts

Since $V_\pi(s) = \sum_a \pi(a;s)\, Q_\pi(s,a)$, the policy advantage needs only Q_π:

$$\mathbb{A}_{\pi,\mu}(\pi') \;=\; \mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\sum_a \big(\pi'(a;s) - \pi(a;s)\big)\, Q_\pi(s,a)\Big]$$

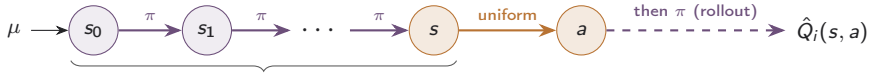


stop at each step w.p. $1-\gamma \Rightarrow s \sim d_{\pi,\mu}$ ▸ Why?

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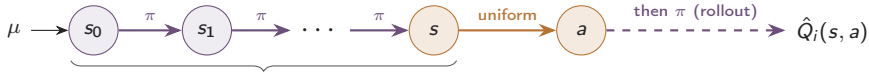


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$$x_i = n_a\, \hat{Q}_i(s,a)\big(\pi'(a;s) - \pi(a;s)\big) \in [-n_a R,\, n_a R], \qquad \hat{\mathbb{A}} = \tfrac{1}{k}\sum_{i=1}^{k} x_i$$

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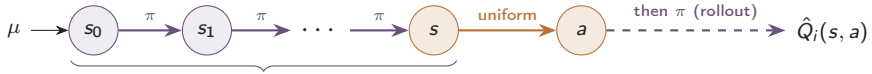
- Unbiased: a uniform $\Rightarrow \mathbb{E}_a\big[n_a f(a)\big] = \sum_a f(a)$ (importance weight n_a, [2]) ▸ IS ▸ Hoeffding

[2] n_a is the number of actions.

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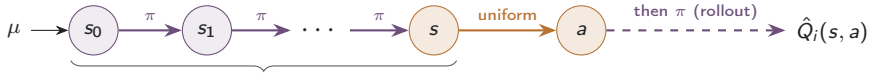
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- Unbiased: a uniform $\Rightarrow \mathbb{E}_a\big[n_a f(a)\big] = \sum_a f(a)$ (importance weight n_a, ▸ IS)[2] ▸ Hoeffding
- Only rollouts of the **current** π — no need to run π_{new}. This is **Q2**.
- CPI: $\Delta = \varepsilon/3$, so $\hat{\mathbb{A}} - \frac{\varepsilon}{3} \le \mathbb{A}$ is safe for the stop test and for α.

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How Good Is the Policy Found?

Theorem 6.2. If $\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) < \varepsilon$, then for any $\tilde{\mu}$:

$$\eta_{\tilde{\mu}}(\pi^*) - \eta_{\tilde{\mu}}(\pi) \;\le\; \frac{\varepsilon}{1-\gamma} \left\| \frac{d_{\pi^*,\tilde{\mu}}}{d_{\pi,\mu}} \right\|_\infty \;\le\; \frac{\varepsilon}{(1-\gamma)^2} \left\| \frac{d_{\pi^*,\tilde{\mu}}}{\mu} \right\|_\infty$$

- **Covariate-shift bound:** test risk $\le$ density ratio $\times$ train risk.
  - train risk: $\mathrm{OPT}(\mathbb{A}_{\pi,\mu})$, on CPI's data $d_{\pi,\mu}$
  - test: the states π^* visits, $d_{\pi^*,\tilde{\mu}}$
- **Uniform μ caps the ratio**: $d_{\pi,\mu} \ge (1-\gamma)\mu \Rightarrow$ ratio $\le |S|/(1-\gamma)$, for any π^*.

CPI: Answers to the Three Questions

1. **Improving measure:** η_μ improves at every update (Thm. 4.4).
2. **Verification:** estimate the policy advantage $\mathbb{A}$ from samples (Hoeffding).
3. **Performance:** within $O(R^2/\varepsilon^2)$ updates, close to optimal up to the mismatch factor (Thm. 6.2).

Kept going forward

- Surrogate on the old state distribution
- Improvement lower bound = gain − penalty
- Small, controlled steps

Limitations

- Only **mixture** policies
- Needs a restart distribution
- Theoretical step sizes are tiny

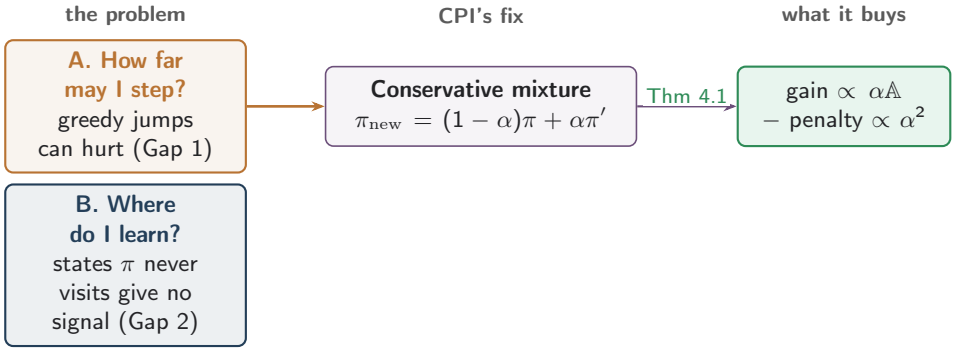
Like non-convex SGD, but Thm. 6.2 turns "stationary" into "near-optimal".

▸ CPI vs. SGD

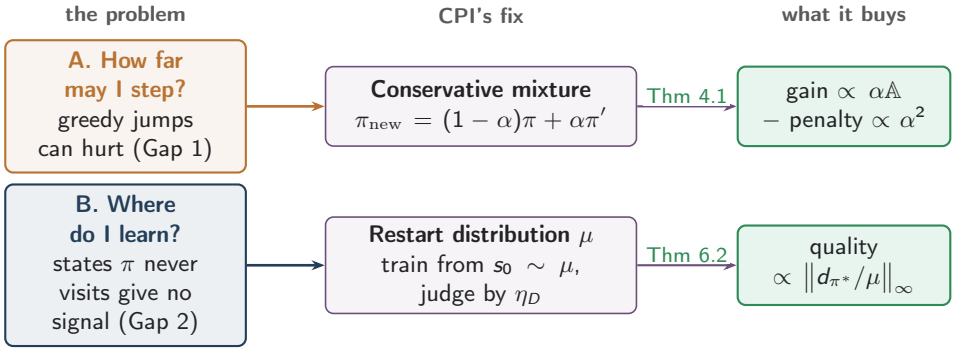
Recap: Two Problems, Two Fixes

the problem	CPI's fix	what it buys
A. How far may I step? greedy jumps can hurt (Gap 1)		
B. Where do I learn? states π never visits give no signal (Gap 2)		

Recap: Two Problems, Two Fixes



Recap: Two Problems, Two Fixes



A New Quantity: the Policy Advantage

$$\mathbb{A}_{\pi,\mu}(\pi') \;=\; \mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\mathbb{E}_{a\sim\pi'(\cdot;s)}\big[A_\pi(s,a)\big]\Big] \;=\; \mathbb{E}_{s\sim d_{\pi,\mu}}\big[\bar{A}(s)\big]$$

- Lemma 6.1's average, but **on the states π visits** (from μ) — so it is **measurable**.
- One step of policy improvement maximizes it.
- It is exactly the **first-order** change of η_μ in α:

▸ Proof

$$\left.\frac{\partial \eta_\mu}{\partial \alpha}\right|_{\alpha=0} = \frac{1}{1-\gamma}\mathbb{A}_{\pi,\mu}(\pi') \qquad \Longrightarrow \qquad \Delta\eta_\mu = \frac{\alpha}{1-\gamma}\mathbb{A}_{\pi,\mu}(\pi') + O(\alpha^2)$$

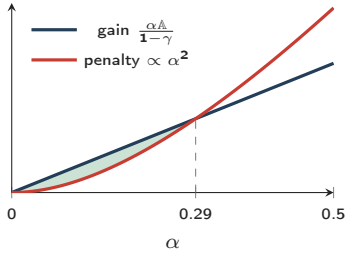
Small α + positive policy advantage $\Rightarrow$ improvement. Large $\alpha \Rightarrow$ possible degradation. How large can α be?

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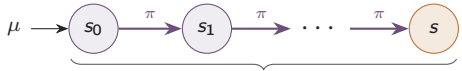
Illustrative: $\gamma = 0.6$, $\varepsilon = 1$, $\mathbb{A} = 0.6$. Green: improvement guaranteed.

Some small step always helps when $\mathbb{A} > 0$. **Which** α? $\rightarrow$ next slide.

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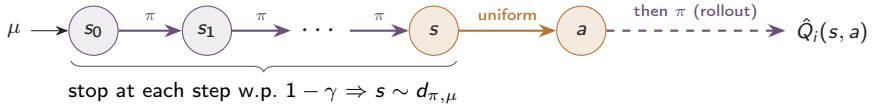
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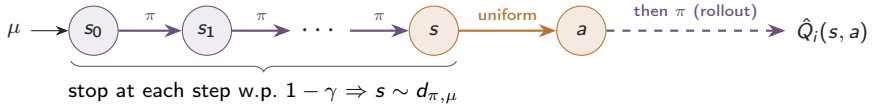
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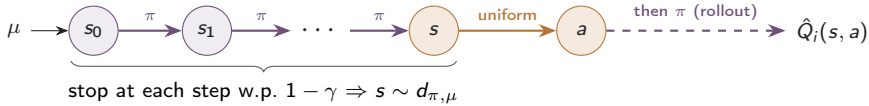
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3. If $\hat{\mathbb{A}} < \frac{2\varepsilon}{3}$: **stop**, return π
4. Else: mix with $\alpha = \frac{(1-\gamma)(\hat{\mathbb{A}} - \varepsilon/3)}{4R}$, go to 1

G_ε: returns π' with $\mathbb{A}_{\pi,\mu}(\pi') \geq \max_{\pi''} \mathbb{A}_{\pi,\mu}(\pi'') - \varepsilon$ — e.g. a regression on advantages with *average* error $\leq \varepsilon/2$.

▸ How?

Theorem 4.4. With prob. $\geq 1 - \delta$, CPI:

i) improves η_μ at **every** update;

ii) stops within $72\, R^2/\varepsilon^2$ calls to G_ε;

iii) returns π with $\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) < 2\varepsilon$.[1]

Number of updates does **not** depend on μ or on the state-space size.

[1] $\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) \equiv \max_{\pi'} \mathbb{A}_{\pi,\mu}(\pi') = \mathbb{E}_{s\sim d_{\pi,\mu}}\big[\max_a A_\pi(s,a)\big]$: the largest policy advantage any π' can achieve (Def. 4.3). Small $\Rightarrow$ no policy improves much on π under $d_{\pi,\mu}$.

CPI: Answers to the Three Questions

1. **Improving measure:** η_μ improves at every update (Thm. 4.4).
2. **Verification:** estimate the policy advantage $\mathbb{A}$ from samples (Hoeffding).
3. **Performance:** within $O(R^2/\varepsilon^2)$ updates, close to optimal up to the mismatch factor (Thm. 6.2).

Kept going forward

- Surrogate on the old state distribution
- Improvement lower bound = gain − penalty
- Small, controlled steps

Limitations

- Only **mixture** policies
- Needs a restart distribution
- Theoretical step sizes are tiny

Like non-convex SGD, but Thm. 6.2 turns "stationary" into "near-optimal".

▸ CPI vs. SGD

Trust Region Policy Optimization

Schulman, Levine, Moritz, Jordan & Abbeel
ICML 2015

Introduces **TRPO**: CPI's guarantee for **any** stochastic policy π_θ

Setting & Assumptions

An MDP $(\mathcal{S}, \mathcal{A}, P, r, \rho_0, \gamma)$:

- **Finite** $\mathcal{S}, \mathcal{A}$ in the proofs — also holds for **continuous** (conditions later).
- **Discounted**: $\gamma \in (0, 1)$, reward $r(s)$.
- **Unknown dynamics**: sample trajectories only.
- **Start distribution** ρ_0 (CPI's D) — **no restart** μ.
- **Unnormalized**: $\eta(\pi) = \mathbb{E}\big[\sum_t \gamma^t r(s_t)\big]$, $s_0 \sim \rho_0$.
- **Any** differentiable stochastic policy π_θ (e.g. a neural net).

From CPI to TRPO

Goal: generalize CPI's guarantee from **mixture policies** to **any** differentiable π_θ.

CPI

- Mixtures: $\pi_\alpha = (1-\alpha)\pi + \alpha\pi'$
- Step size: $\alpha \in [0, 1]$
- New policy: greedy chooser G_ε

TRPO

- **Any** π_θ (e.g. neural net)
- Step: $\theta \to \theta + \Delta\theta$
- How far? Need a **distance**

Strategy: start from the same performance difference identity.

The Performance Difference Identity

Unnormalized: $\eta(\pi) = \mathbb{E}\big[\sum_t \gamma^t r(s_t)\big]$, $\rho_\pi(s) = \sum_t \gamma^t P(s_t = s)$.

$$\eta(\tilde{\pi}) = \eta(\pi) + \sum_s \rho_{\tilde{\pi}}(s) \sum_a \tilde{\pi}(a \mid s)\, A_\pi(s, a)$$

The problem: $\rho_{\tilde{\pi}}$ depends on $\tilde{\pi}$ in a **complicated** way.

- Changes in $\tilde{\pi}$ affect which states are visited
- This couples all the sums together

Solution: use a **local surrogate** with the old state distribution.

The Local Surrogate

Key idea: replace the **new** visitation $\rho_{\tilde{\pi}}$ with the **old** visitation ρ_π.

$$L_\pi(\tilde{\pi}) \;=\; \eta(\pi) + \sum_s \rho_\pi(s) \sum_a \tilde{\pi}(a \mid s)\, A_\pi(s, a)$$

- Computable from data collected under π
- Uses the **old** state distribution ρ_π, which we know
- Same construction as CPI's policy advantage

Question: how good is this approximation?

A First-Order Approximation

$L_\pi(\tilde{\pi})$ matches $\eta(\tilde{\pi})$ **to first order** around π.

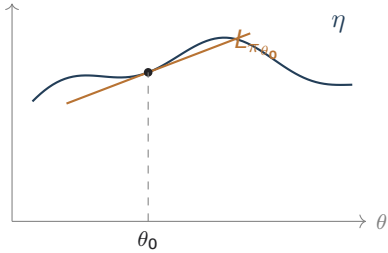
For any parameterization π_θ, at θ_0:

1. Equal value:

$$L_{\pi_{\theta_0}}(\pi_{\theta_0}) = \eta(\pi_{\theta_0})$$

2. Equal gradient:

$$\nabla_\theta L_{\pi_{\theta_0}}(\pi_\theta)\big|_{\theta=\theta_0} = \nabla_\theta \eta(\pi_\theta)\big|_{\theta=\theta_0}$$



Tangent line at θ_0

To know **how far** we can step, we need a **bound** on $|\eta(\tilde{\pi}) - L_\pi(\tilde{\pi})|$.

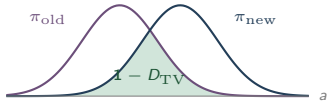
Measuring Policy Distance

To bound $|\eta(\tilde{\pi}) - L_\pi(\tilde{\pi})|$, we need a **distance** between policies.

Total variation distance:

$$D_{\mathrm{TV}}(p \,\|\, q) = \tfrac{1}{2} \sum_a \left| p(a) - q(a) \right|$$

- Range: $[0, 1]$
- Interpretation: $1 - D_{\mathrm{TV}}$ = probability that optimally coupled samples agree



Shared mass

Policy distance: maximum over all states.

$$\alpha = D_{\mathrm{TV}}^{\max}(\pi_{\mathrm{old}}, \pi_{\mathrm{new}}) = \max_s D_{\mathrm{TV}}\big(\pi_{\mathrm{old}}(\cdot \mid s) \,\|\, \pi_{\mathrm{new}}(\cdot \mid s)\big)$$

Theorem 1: Bounding the Surrogate Error

Theorem 1 (TRPO's main guarantee):
Let $\alpha = D_{\mathrm{TV}}^{\max}(\pi_{\mathrm{old}}, \pi_{\mathrm{new}})$ and $\epsilon = \max_{s,a} |A_\pi(s,a)|$, then

$$\eta(\pi_{\mathrm{new}}) \;\geq\; L_{\pi_{\mathrm{old}}}(\pi_{\mathrm{new}}) - \frac{4\epsilon\gamma}{(1-\gamma)^2}\,\alpha^2$$

Connection to CPI:

- **Same structure:** $\eta \geq L - C\alpha^2$ (quadratic penalty)
- **Key change:** mixing weight $\alpha \in [0,1] \to$ policy distance $D_{\mathrm{TV}}^{\max}(\pi_{\mathrm{old}}, \pi_{\mathrm{new}})$
- **Slightly looser:** $\epsilon = \max_{s,a} |A_\pi(s,a)|$ instead of CPI's $\max_s \left|\mathbb{E}_{a\sim\pi'}[A_\pi(s,a)]\right|$
- **Gain:** works for **any** π_θ, not just mixtures

▸ Continuous S, A?

From Total Variation to KL Divergence

Problem
TV distance D_{TV} isn't smooth in θ.

Solution
Use KL divergence instead.

Pinsker's inequality: $D_{\mathrm{TV}}(p\|q)^2 \le D_{\mathrm{KL}}(p\|q) \quad \left(:= \sum_a p(a) \log \frac{p(a)}{q(a)}\right)$

Substituting into Theorem 1: (for $D_{\mathrm{KL}}^{\max}(\pi, \tilde{\pi}) = \max_s D_{\mathrm{KL}}\big(\pi(\cdot|s) \,\|\, \tilde{\pi}(\cdot|s)\big)$)

$$\eta(\tilde{\pi}) \ge L_\pi(\tilde{\pi}) - C\, D_{\mathrm{KL}}^{\max}(\pi, \tilde{\pi}), \qquad C = \frac{4\epsilon\gamma}{(1-\gamma)^2}$$

✓ KL is smooth, differentiable in θ ✓ Still a valid lower bound on η

The Minorization–Maximization Algorithm

Define the lower bound: $M_i(\tilde{\pi}) = L_{\pi_i}(\tilde{\pi}) - C\, D_{\mathrm{KL}}^{\max}(\pi_i, \tilde{\pi})$

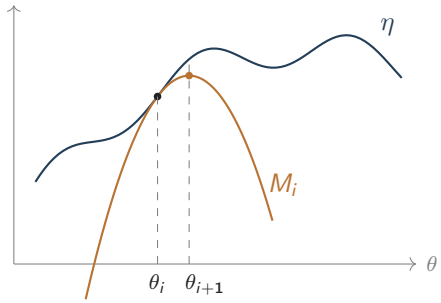
Properties:

- $M_i(\tilde{\pi}) \leq \eta(\tilde{\pi})$ everywhere
- $M_i(\pi_i) = \eta(\pi_i)$ (tangent)

Algorithm: maximize M_i to get π_{i+1}. Then:

$$\eta(\pi_{i+1}) \geq M_i(\pi_{i+1}) \geq M_i(\pi_i) = \eta(\pi_i)$$

Monotone improvement!



This is what **the theory guarantees**. Next: making it practical.

From Penalty to Constraint: the Trust Region

Theory (penalty)

$$\max_{\theta} \; L_{\theta_{\text{old}}}(\theta) - C\, D_{\text{KL}}^{\max}(\theta_{\text{old}}, \theta)$$

With the theoretical C, steps are **tiny**; and a good C is hard to choose robustly.

Practice (constraint)

$$\max_{\theta} \; L_{\theta_{\text{old}}}(\theta)$$

$$\text{s.t.} \;\; D_{\text{KL}}^{\max}(\theta_{\text{old}}, \theta) \le \delta$$

δ is in policy space, so it transfers across problems ($\delta = 0.01$ in all experiments).

Why needed

$C = \frac{4\epsilon\gamma}{(1-\gamma)^2}$: at $\gamma = 0.99$, $C \approx 4 \cdot 10^4\, \epsilon \Rightarrow$ tiny steps.

What changes

A hard limit δ on the KL replaces the penalty C: we optimize L where we *trust* it.

What it costs

The guarantee becomes a heuristic; δ is a hyperparameter.

From Max KL to Mean KL

1. penalty → constraint → **2. max → mean KL** → 3. samples + IS → 4. linear / quadratic → 5. CG + line search

Replace the max over states by the average under the old visitation distribution:

$$\bar{D}_{\mathrm{KL}}^{\rho}(\theta_1, \theta_2) = \mathbb{E}_{s\sim\rho}\Big[D_{\mathrm{KL}}\big(\pi_{\theta_1}(\cdot \mid s) \,\|\, \pi_{\theta_2}(\cdot \mid s)\big)\Big]$$

$$\max_{\theta} \; L_{\theta_{\mathrm{old}}}(\theta) \quad \text{s.t.} \quad \bar{D}_{\mathrm{KL}}^{\rho_{\theta_{\mathrm{old}}}}(\theta_{\mathrm{old}}, \theta) \le \delta$$

Why needed

$D_{\mathrm{KL}}^{\max}$ is one constraint **per state**, also at states we never sample.

What changes

An **expectation** over visited states $\Rightarrow$ estimated from the same rollouts.

What it costs

Rarely visited states may change a lot. Empirically like max KL (results at the end).

Turning Sums Into Samples

Starting point: The local surrogate objective

$$\max_{\theta} \; L_{\theta_{\text{old}}}(\theta) = \sum_{s} \rho_{\theta_{\text{old}}}(s) \sum_{a} \pi_{\theta}(a \mid s)\, A_{\theta_{\text{old}}}(s, a)$$

Three transformations:

1. Replace $\sum_s$ with expectation $\mathbb{E}_{s \sim \rho_{\theta_{\text{old}}}}$ (▸ sample states from rollouts)
2. Replace A with Q (▸ constant shift)
3. Replace $\sum_a$ with ▸ importance sampling (reweight actions)

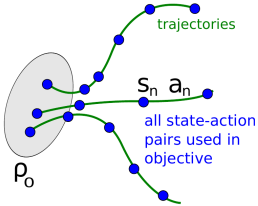
Practical objective:

$$\max_{\theta} \; \mathbb{E}_{s \sim \rho_{\theta_{\text{old}}},\, a \sim q}\left[\frac{\pi_{\theta}(a \mid s)}{q(a \mid s)}\, Q_{\theta_{\text{old}}}(s, a)\right]$$

$$\text{s.t.} \;\; \mathbb{E}_{s \sim \rho_{\theta_{\text{old}}}}\left[D_{\text{KL}}\left(\pi_{\theta_{\text{old}}}(\cdot \mid s) \,\|\, \pi_{\theta}(\cdot \mid s)\right)\right] \le \delta$$

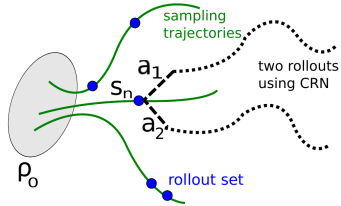
Two Sampling Schemes

TRPO offers two ways to collect data and estimate the objective:



Single path

Roll out $\pi_{\theta_{\text{old}}}$ and use **every** (s, a) pair encountered along the trajectory.



Vine

From selected states in a trunk rollout, branch out: try multiple actions and run short rollouts.

Single Path vs Vine: How They Work

Single path

- Sampling distribution: $q = \pi_{\theta_{\text{old}}}$
- $\hat{Q}$ from discounted return-to-go
- Uses **every** state-action pair
- **No resets needed**

When: Real systems, robotics — anywhere you cannot reset to arbitrary states.

Vine

- From each state in rollout set: try K actions $a_k \sim q$
- Short rollout per action (CRN)
- **Lower variance** $\hat{Q}$
- Needs **state resets**

When: Simulators with reset capability — resets only to states from ρ_0 rollouts (not arbitrary μ like CPI).

Key limitation both share: Only actions are reweighted by $\pi_\theta/\pi_{\theta_{\text{old}}}$. States still come from $s \sim \rho_{\theta_{\text{old}}}$ (cannot reweight states without knowing $\rho_\theta/\rho_{\theta_{\text{old}}}$).

Solving the Trust-Region Problem: Search Direction

Approximate around θ_{old} — **linear** objective, **quadratic** constraint:

$$L(\theta) \approx L(\theta_{\text{old}}) + g^\top(\theta - \theta_{\text{old}}), \qquad \bar{D}_{\text{KL}}(\theta_{\text{old}}, \theta) \approx \tfrac{1}{2}(\theta - \theta_{\text{old}})^\top A\,(\theta - \theta_{\text{old}})$$

- $A_{ij} = \frac{\partial^2}{\partial\theta_i \partial\theta_j}\bar{D}_{\text{KL}}(\theta_{\text{old}}, \theta)$: the **curvature of the KL**, a.k.a. the **Fisher information matrix**.
- Solution direction: $s \propto A^{-1}g$ — the **natural gradient**: steepest ascent when distance is measured in KL, not in $\|\theta\|$.

Why needed
L and the KL are nonlinear in θ; an exact constrained solve per batch is too expensive.

What changes
First/second-order Taylor models $\Rightarrow$ a closed-form direction $A^{-1}g$.

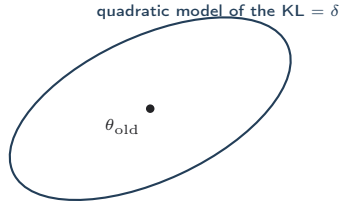
What it costs
Valid only locally — the step must be checked (next slide).

Computing the Step: Conjugate Gradient + Line Search

1. penalty → constraint → 2. max → mean KL → 3. samples + IS → 4. linear / quadratic → **5. CG + line search**

Direction: solve $As = g$ by **conjugate gradient**

only needs products $y \mapsto Ay$ (autodiff), $k = 10$ iterations



A has $|\theta|^2$ entries — never formed.

Why needed
A cannot even be stored for a neural net; the quadratic model is only local.

What changes
CG with Fisher-vector products; full step to δ; line search on the true L and KL.

What it costs
Second order and **one step per batch** — PPO's starting point.

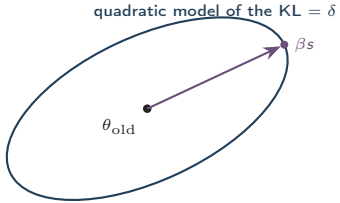
Computing the Step: Conjugate Gradient + Line Search

Direction: solve $As = g$ by **conjugate gradient**

only needs products $y \mapsto Ay$ (autodiff), $k = 10$ iterations

Largest step that reaches the KL boundary

$$\tfrac{1}{2}(\beta s)^\top A(\beta s) = \delta \;\Rightarrow\; \beta = \sqrt{2\delta / s^\top As}$$



Step to the edge of the *model*'s trust region.

Why needed
A cannot even be stored for a neural net; the quadratic model is only local.

What changes
CG with Fisher-vector products; full step to δ; line search on the true L and KL.

What it costs
Second order and **one step per batch** — PPO's starting point.

Computing the Step: Conjugate Gradient + Line Search

Direction: solve $As = g$ by **conjugate gradient**

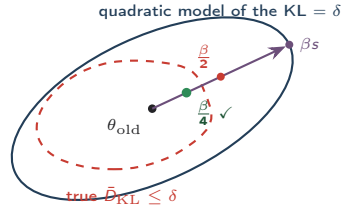
only needs products $y \mapsto Ay$ (autodiff), $k = 10$ iterations

Largest step that reaches the KL boundary

$$\tfrac{1}{2}(\beta s)^\top A(\beta s) = \delta \;\Rightarrow\; \beta = \sqrt{2\delta / s^\top As}$$

Backtrack $\beta,\ \beta/2,\ \beta/4, \ldots$ until

$L_{\theta_{\text{old}}}(\theta)$ improves and $\bar{D}_{\text{KL}}(\theta_{\text{old}}, \theta) \le \delta$



The model is only local: shrink until the **true** checks pass.

Why needed

A cannot even be stored for a neural net; the quadratic model is only local.

What changes

CG with Fisher-vector products; full step to δ; line search on the true L and KL.

What it costs

Second order and **one step per batch** — PPO's starting point.

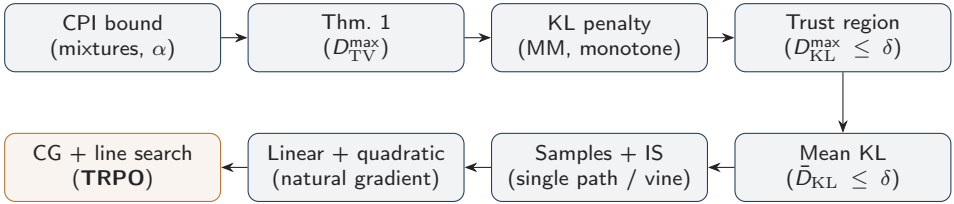
The TRPO Algorithm

Repeat:

1. Collect (s, a) pairs with Monte Carlo $\hat{Q}$ estimates (single path or vine).
2. Build the sample estimates of the surrogate L and the mean-KL constraint.
3. Compute $g = \nabla_\theta L$; solve $As = g$ with conjugate gradient (Fisher-vector products).
4. Step size $\beta = \sqrt{2\delta / s^\top As}$; backtracking line search on L and the KL.
5. $\theta_{\text{old}} \leftarrow \theta_{\text{old}} + \beta s$.

Cost per update: only slightly more than computing the gradient itself.

Recap: From Theory to Practice



- First three boxes: **theory** (monotone improvement). From the trust region on: **practice** — each arrow trades a guarantee for practicality.
- What's left: noisy $\hat{Q}$ (GAE) and a heavy second-order solver (PPO).

▸ Related algorithms

GAE and PPO — Main Sources



Published parallel to TRPO, this introduced the exponentially-weighted advantage estimator used inside almost all modern PPO and TRPO implementations to balance variance and bias.

$$\hat{A}_t^{GAE(\gamma,\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l (r_{t+l} + \gamma V(s_{t+l+1}) - V(s_{t+l}))$$



Simplified TRPO by replacing the expensive KL penalty with a clipped surrogate objective function. This became the industry standard for Reinforcement Learning from Human Feedback (RLHF).

Where TRPO Leaves Us

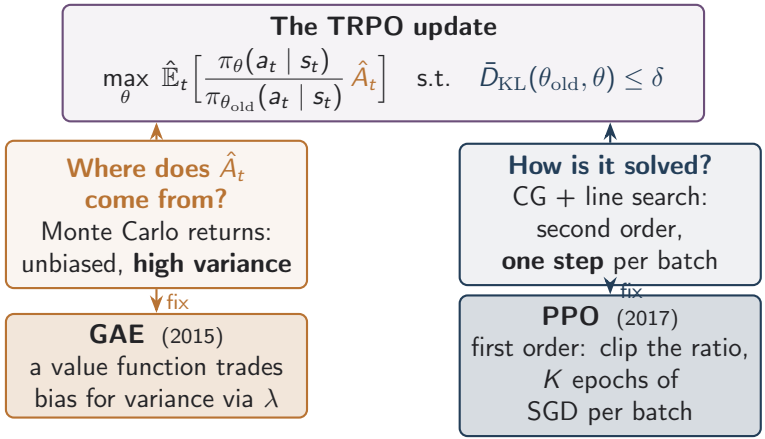
The TRPO update

$$\max_{\theta} \hat{\mathbb{E}}_t \left[\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)} \hat{A}_t \right] \quad \text{s.t.} \quad \bar{D}_{\text{KL}}(\theta_{\text{old}}, \theta) \leq \delta$$

Where does $\hat{A}_t$ come from?
Monte Carlo returns: unbiased, **high variance**

How is it solved?
CG + line search: second order, **one step** per batch

Where TRPO Leaves Us



High-Dimensional Continuous Control Using Generalized Advantage Estimation

Schulman, Moritz, Levine, Jordan & Abbeel
ICLR 2016 (arXiv 2015)

Introduces **GAE**(γ, λ): TD(λ) for advantages

Key Contributions

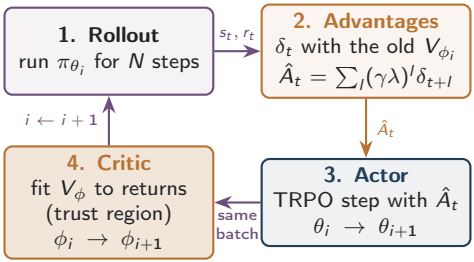
- **Solves TRPO's variance problem:** replace Monte Carlo returns with bootstrapped advantage estimates.
- **Actor–critic architecture:** learn a value function $V_\phi(s)$ alongside the policy.
- λ **trades bias for variance:** $\lambda = 0$ (pure TD) has low variance, high bias; $\lambda = 1$ (Monte Carlo) is unbiased but high variance.
- **GAE**(γ, λ)**:** exponentially weighted average of n-step TD advantages, $\hat{A}_t = \sum_l (\gamma\lambda)^l \delta_{t+l}$, with the **TD residual** $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ (itself a one-step advantage estimate).

High-Dimensional Continuous Control Using Generalized Advantage Estimation

Schulman, Moritz, Levine, Jordan & Abbeel
ICLR 2016 (arXiv 2015)

Introduces **GAE**(γ, λ): TD(λ) for advantages

One Iteration: Actor, then Critic



Order matters: $\hat{A}_t$ uses the *old* critic. Fitting V first can overfit the batch $\Rightarrow$ every $\delta_t \approx 0 \Rightarrow$ zero gradient.

Proximal Policy Optimization Algorithms

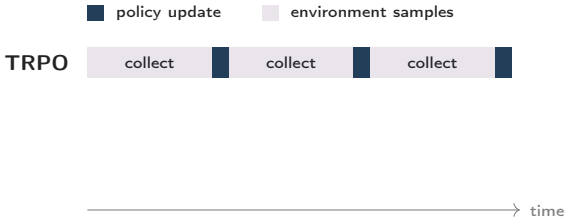
Schulman, Wolski, Dhariwal, Radford & Klimov
arXiv 2017 (OpenAI)

Introduces **PPO**: TRPO's reliability with only **first-order** SGD

Setting & Motivation

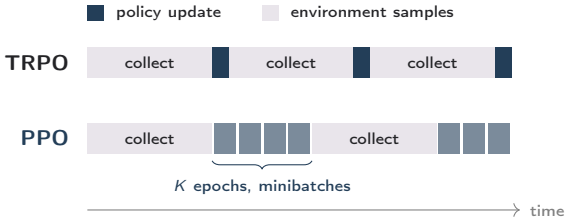
- Same setting as TRPO; advantages $\hat{A}_t$ from (truncated) **GAE**.
- **TRPO**: complicated; no dropout, no parameter sharing between policy and value.
- **Vanilla PG**: several steps on the same batch $\Rightarrow$ destructively large updates.
- **Goal**: reuse each batch for many SGD steps, without leaving the trust region.
- Tested on MuJoCo and 49 Atari games.

PPO's Goal: Many SGD Steps per Batch



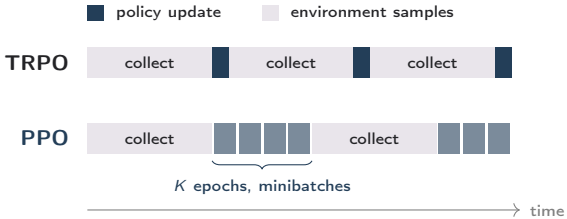
TRPO: one (expensive) step, then the batch is thrown away.

PPO's Goal: Many SGD Steps per Batch

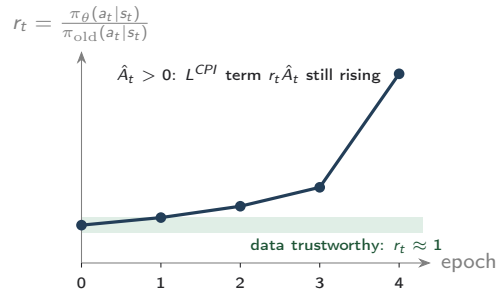


PPO: squeeze the same batch with cheap first-order steps.

PPO's Goal: Many SGD Steps per Batch

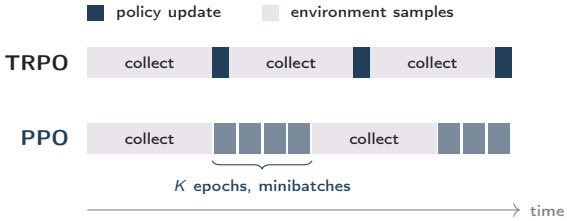


PPO: squeeze the same batch with cheap first-order steps.

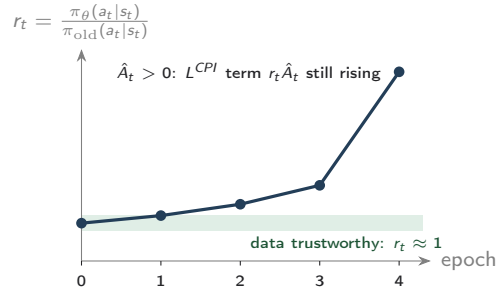


One sample, $\hat{A}_t > 0$: **nothing says stop**.

PPO's Goal: Many SGD Steps per Batch



PPO: squeeze the same batch with cheap first-order steps.



One sample, $\hat{A}_t > 0$: **nothing says stop**.

The data came from π_{old}: the surrogate is trustworthy only near $r_t = 1$. Unchecked, repeated steps give **destructively large** updates.

Back to the CPI Surrogate

Ratio = TRPO's importance weight: $r_t(\theta) = \dfrac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}, \quad r_t(\theta_{\text{old}}) = 1$

$$L^{CPI}(\theta) = \hat{\mathbb{E}}_t\left[r_t(\theta)\,\hat{A}_t\right] \qquad \text{("CPI": Kakade \& Langford)}$$

Back to the CPI Surrogate

Ratio = TRPO's importance weight: $r_t(\theta) = \dfrac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}, \quad r_t(\theta_{\text{old}}) = 1$

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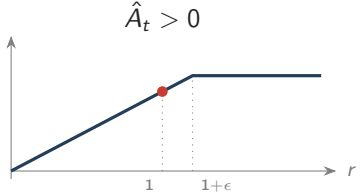
Constraint (TRPO)
s.t. $\hat{\mathbb{E}}_t[D_{\text{KL}}] \le \delta$
second-order solver

Penalty (baseline)
$-\beta\,\hat{\mathbb{E}}_t[D_{\text{KL}}]$
no single β works $\rightarrow$ adapt it

Clip ("PPO")
no gain once $|r_t - 1| > \epsilon$
first order, next slide

The Clipped Surrogate Objective

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t\Big[\min\left(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\,\hat{A}_t\right)\Big], \qquad \epsilon = 0.2$$

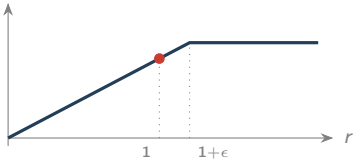


Raise $\pi(a_t|s_t)$ — no extra gain past $1+\epsilon$.

The Clipped Surrogate Objective

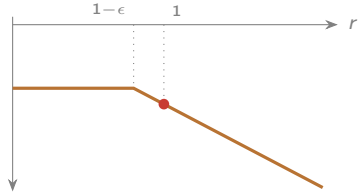
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$\hat{A}_t > 0$



Raise $\pi(a_t|s_t)$ — no extra gain past $1+\epsilon$.

$\hat{A}_t < 0$

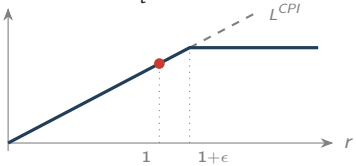


Lower $\pi(a_t|s_t)$ — no extra gain below $1-\epsilon$.

The Clipped Surrogate Objective

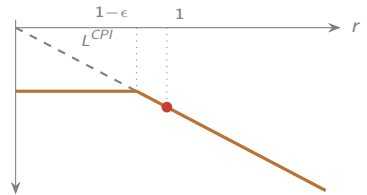
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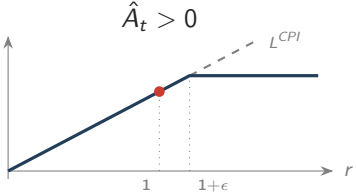


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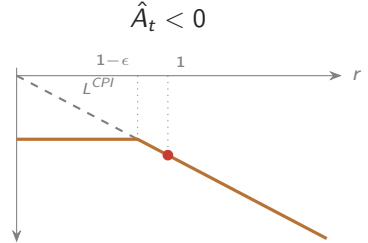
min $\Rightarrow$ $L^{CLIP} \leq L^{CPI}$: clipping removes **gains**, never hides **drops** — a pessimistic bound.

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Raise $\pi(a_t|s_t)$ — no extra gain past $1+\epsilon$.
Wrong way ($r < 1-\epsilon$): not clipped, objective keeps falling.



Lower $\pi(a_t|s_t)$ — no extra gain below $1-\epsilon$.
Wrong way ($r > 1+\epsilon$): not clipped, objective keeps falling.

min $\Rightarrow$ $L^{CLIP} \le L^{CPI}$: clipping removes **gains**, never hides **drops** — a pessimistic bound.
Moving the wrong way keeps its gradient: the update can always pull r_t **back** toward 1.

What Clipping Replaces

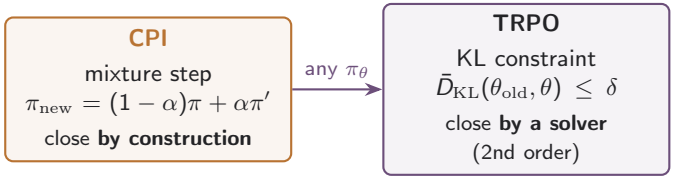
CPI

mixture step

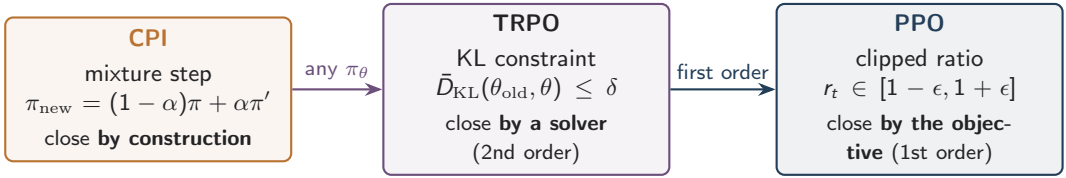
$\pi_{\text{new}} = (1 - \alpha)\pi + \alpha\pi'$

close **by construction**

What Clipping Replaces

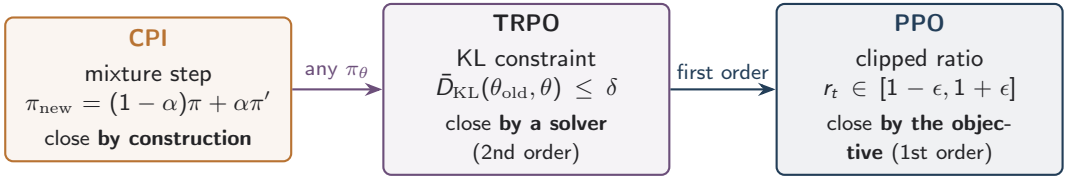


What Clipping Replaces



Proximal: keep π_θ near π_{old} — with plain minibatch SGD/Adam, several epochs per batch.

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Clipping removes the **incentive** to move a ratio further — it does not **enforce** a limit. No KL bound is guaranteed.

PPO in Practice

$$L_t^{CLIP+VF+S}(\theta) = \hat{\mathbb{E}}_t\Big[\underbrace{L_t^{CLIP}(\theta)}_{\text{policy}} - c_1\underbrace{\big(V_\theta(s_t) - V_t^{\text{targ}}\big)^2}_{\text{value}} + c_2\underbrace{S[\pi_\theta](s_t)}_{\text{entropy}}\Big]$$

Each iteration:

1. N actors run $\pi_{\theta_{\text{old}}}$ for T steps each.
2. Truncated GAE: $\hat{A}_t = \sum_{l=0}^{T-t-1}(\gamma\lambda)^l\delta_{t+l}$.
3. K epochs of minibatch Adam on L.
4. $\theta_{\text{old}} \leftarrow \theta$.

The point: K epochs on the **same** batch.

TRPO took **one** step per batch; clipping is what makes reuse safe.

$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$: TD residual, as in GAE.

TRPO vs. PPO

	TRPO	PPO
Stay close by	KL constraint $\bar{D}_{\mathrm{KL}} \leq \delta$	clipped ratio $[1-\epsilon, 1+\epsilon]$
Solver	CG + line search (2nd order)	minibatch Adam (1st order)
Steps per batch	one	K epochs
Policy + value	separate networks	may share (one loss)
Advantages	Monte Carlo $\hat{Q}$	truncated GAE
States from	$s_0 \sim \rho_0$, then π_{old}	same — still no μ

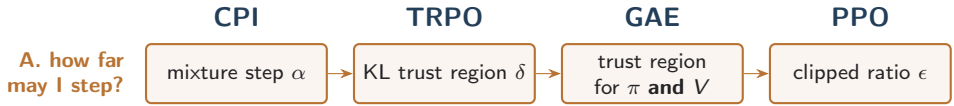
Kept from CPI and TRPO

- Surrogate on the **old** samples
- The ratio $\pi_\theta / \pi_{\theta_{\mathrm{old}}}$
- Small, controlled steps

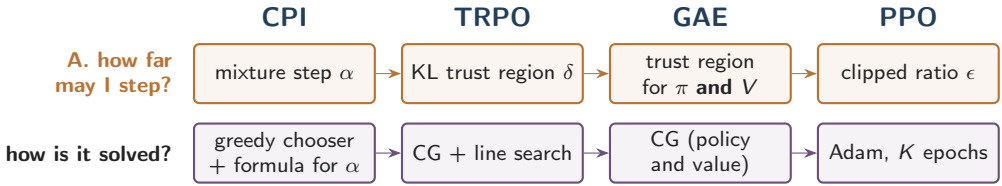
Given up

- Any improvement **bound**
- An exact limit on the KL
- (Monotone improvement: never had it in practice)

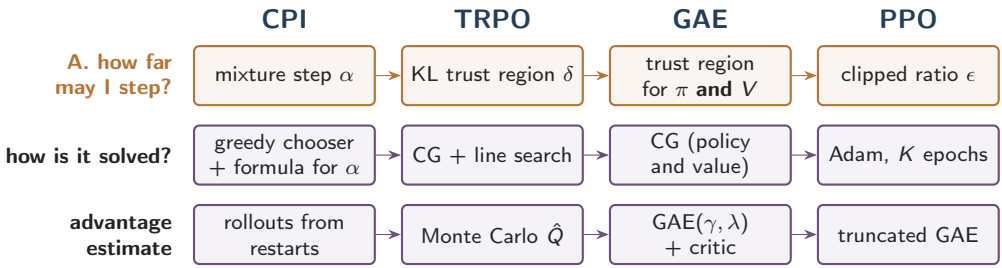
The Whole Story: CPI $\rightarrow$ TRPO $\rightarrow$ GAE $\rightarrow$ PPO



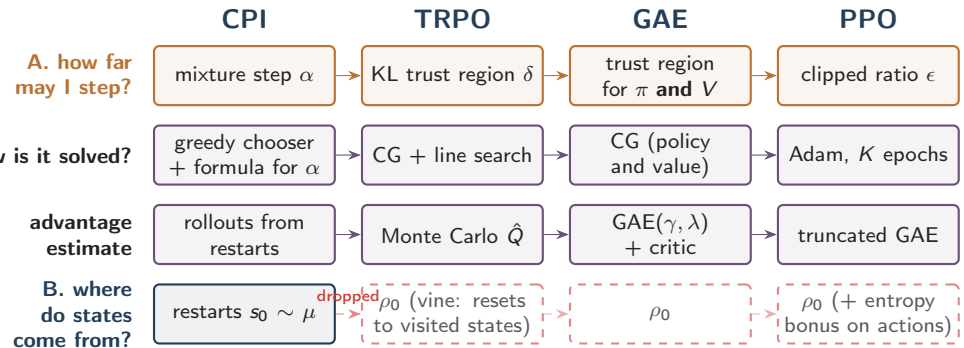
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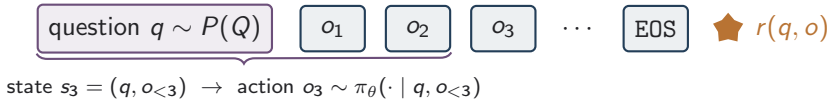
The Whole Story: CPI → TRPO → GAE → PPO



One Line per Paper

Paper	Problem it attacks	Main idea
CPI	greedy updates from approximate values can hurt	conservative mixture, step α from a bound
CPI	on-policy samples miss important states	exploratory restart distribution μ
TRPO	mixtures don't fit neural-net policies	step size $=$ a policy distance (KL)
TRPO (practice)	the theoretical penalty is far too conservative	mean-KL trust region, CG $+$ line search
GAE	$\hat{A}_t$ from returns is too noisy	critic V $+$ bias–variance knob λ
PPO	second-order step, one step per batch	clipped ratio $+$ minibatch SGD, K epochs

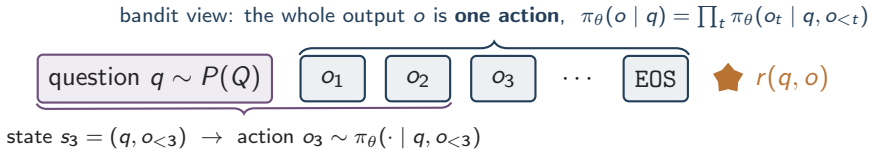
Text Generation as an RL Problem



Start distribution
the questions: $q \sim P(Q)$
plays the role of ρ_0

Dynamics
append the token:
known and
deterministic

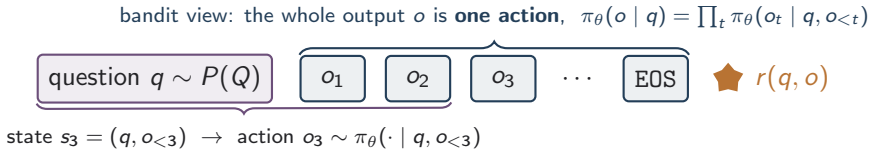
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Reward
a learned **reward model**
scores o — only at the end

DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models

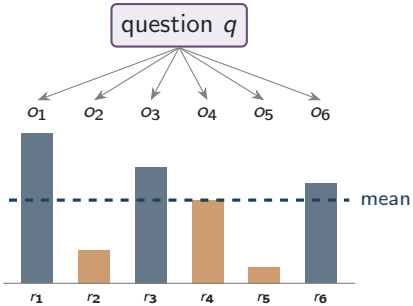
Shao, Wang, Zhu, et al.
(DeepSeek-AI)
arXiv 2024

Introduces **GRPO**: PPO without the critic

Setting & Assumptions

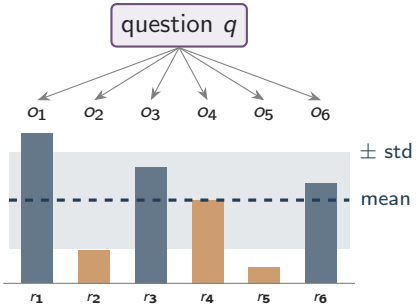
- Math questions q (GSM8K, MATH; chain of thought), policy: DeepSeekMath-Instruct 7B.
- A **reward model** scores each output o — at the end (outcome) or per step (process).
- Online: $G = 64$ outputs per question from the current policy.
- Problem with PPO's critic: **as large as the policy**, and hard to fit per token when the reward comes only at the end.

Group-Relative Advantages



Sample G outputs for the **same** question; score them.

Group-Relative Advantages

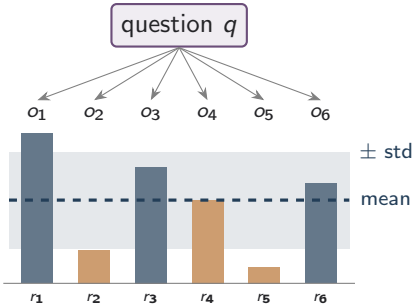


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the same value for every token t of o_i

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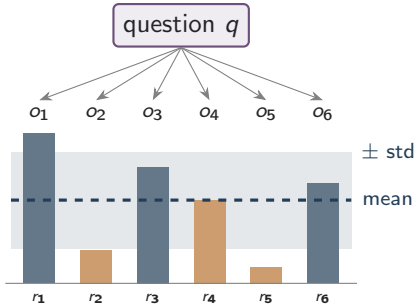
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- Matches how RMs are trained: **comparisons** within one question.

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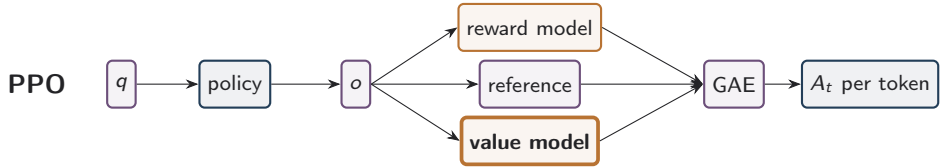
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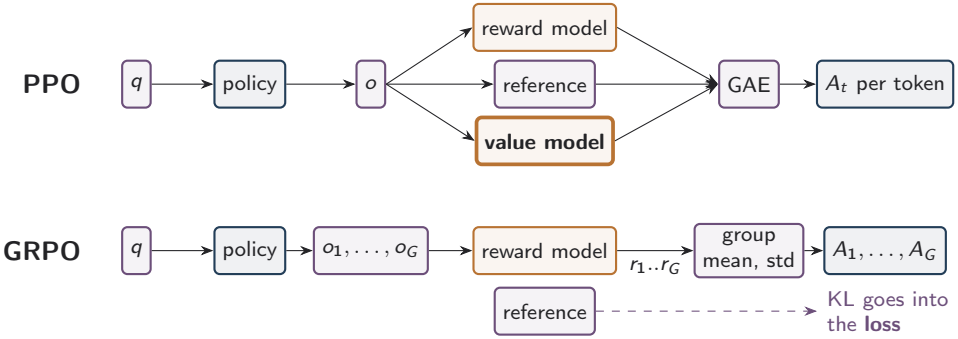
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RL on GSM8K + MATH questions only: GSM8K 82.9 → 88.2%, MATH 46.8 → 51.7%, and gains out of domain.

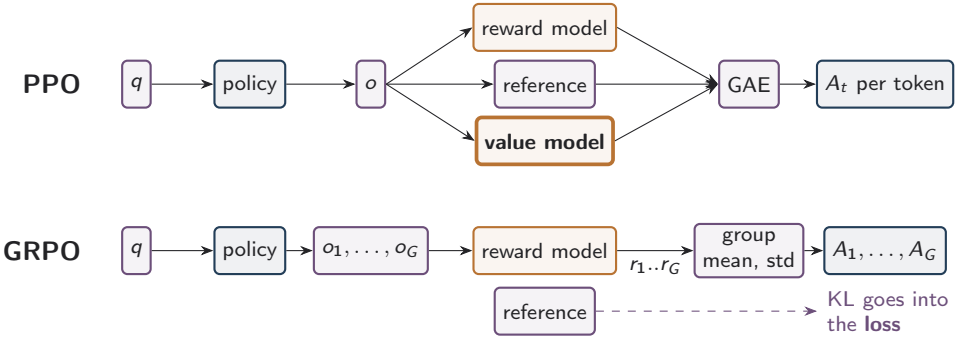
PPO vs. GRPO


PPO
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o
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$o_1, \ldots, o_G$
reward model
$r_1..r_G$
group mean, std
$A_1, \ldots, A_G$
reference
KL goes into the loss

PPO vs. GRPO



$$\mathcal{J}_{\mathrm{GRPO}}(\theta) = \mathbb{E}\Big[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\Big\{\min\left(\rho_{i,t}\hat{A}_{i,t},\ \mathrm{clip}(\rho_{i,t}, 1\pm\varepsilon)\,\hat{A}_{i,t}\right) - \beta\, D_{\mathrm{KL}}\left[\pi_\theta \| \pi_{\mathrm{ref}}\right]\Big\}\Big]$$

$\rho_{i,t} = \pi_\theta(o_{i,t}|q, o_{i,<t}) / \pi_{\theta_{\mathrm{old}}}(o_{i,t}|q, o_{i,<t})$: still PPO's clipped ratio, per token.

GRPO Through the Lens of This Talk

Variance: full circle

GAE learned a critic because Monte Carlo returns were too noisy. GRPO goes **back to Monte Carlo**: G answers per question make a baseline cheap.

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RLOO (Ahmadian et al. 2024)

baseline b_i = mean of the **other** $G-1$ rewards: $r_i - \bar{r} = \frac{G-1}{G}(r_i - b_i)$, the same up to scale. Real differences: GRPO's std and $1/|o_i|$.

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Why the critic could go: samples per question are cheap, and the question is the only real state.

Congratualations ...

You are now a PPO expert !!

Let's go and drink some coffee.

Proof: $\partial \eta_\mu / \partial \alpha$ at $\alpha = 0$ (1/3)

The conservative mixtures form a **one-parameter policy class** $\{\pi_\alpha\}_{\alpha \in [0,1]}$:

$$\pi_\alpha(a;s) = (1-\alpha)\,\pi(a;s) + \alpha\,\pi'(a;s) \quad \Longrightarrow \quad \frac{\partial \pi_\alpha(a;s)}{\partial \alpha} = \pi'(a;s) - \pi(a;s)$$

So the policy gradient formula applies as is — α **plays the role of** θ:

Policy gradient, any $\theta \in \mathbb{R}^m$: $\nabla_\theta \eta_\mu = \dfrac{1}{1-\gamma} \sum_s d_{\pi_\theta,\mu}(s) \sum_a \nabla_\theta \pi_\theta(a;s)\, Q_{\pi_\theta}(s,a)$

$\theta := \alpha$

Mixture, at $\alpha = 0$ $(\pi_0 = \pi)$: $\left.\dfrac{\partial \eta_\mu}{\partial \alpha}\right|_{\alpha=0} = \dfrac{1}{1-\gamma} \sum_s d_{\pi,\mu}(s) \sum_a \dfrac{\partial \pi_\alpha(a;s)}{\partial \alpha}\, Q_\pi(s,a)$

The factor $\frac{1}{1-\gamma}$ comes from our normalization: $d_{\pi,\mu}$ sums to 1 and Q_π carries a $(1-\gamma)$.

Proof: $\partial \eta_\mu / \partial \alpha$ at $\alpha = 0$ (2/3)

Substitute the derivative of the mixture:

$$\left.\frac{\partial \eta_\mu}{\partial \alpha}\right|_{\alpha=0} = \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\sum_a \big(\pi'(a;s) - \pi(a;s)\big)\, Q_\pi(s,a)\Big]$$

Since $\sum_a \pi(a;s)\, Q_\pi(s,a) = V_\pi(s)$, $\sum_a \pi'(a;s) = 1$ and $A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s)$, the inner sum is:

$$\sum_a \pi'(a;s)\, Q_\pi(s,a) - V_\pi(s) = \mathbb{E}_{a\sim\pi'(\cdot;s)}\big[A_\pi(s,a)\big]$$

Back into the expectation — this is the policy advantage $\mathbb{A}_{\pi,\mu}(\pi')$:

$$\left.\frac{\partial \eta_\mu}{\partial \alpha}\right|_{\alpha=0} = \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim d_{\pi,\mu}}\Big[\mathbb{E}_{a\sim\pi'(\cdot;s)}\big[A_\pi(s,a)\big]\Big] = \frac{1}{1-\gamma}\,\mathbb{A}_{\pi,\mu}(\pi')$$

Proof: α and θ Play the Same Role (3/3)

CPI: parameter $\alpha \in [0, 1]$

$$\eta_\mu(\pi_\alpha) = \eta_\mu(\pi) + \frac{\alpha}{1-\gamma}\mathbb{A}_{\pi,\mu}(\pi') + O(\alpha^2)$$

old-state surrogate exact to 1st order (this proof)

step size: α **itself**

generalize

TRPO: parameter $\theta \in \mathbb{R}^m$

$$\eta(\pi_\theta) = L_{\pi_{\theta_0}}(\pi_\theta) + o(\|\theta - \theta_0\|)$$

old-state surrogate exact to 1st order (TRPO Eq. 4)

step size (Thm. 1): $\alpha = D_{\mathrm{TV}}^{\max}(\pi_{\mathrm{old}}, \pi_{\mathrm{new}})$

The proof never used that α is a mixing weight — only that it parameterizes the policy. TRPO keeps the first-order argument for any θ, and replaces "how far" (α) by a distance between policies.

◂ Back to the policy advantage ◂ Back to the policy advantage recap

From $\mathbb{E}_{s\sim d}$ to a Sum Over Time

$$\frac{1}{1-\gamma}\mathbb{E}_{s\sim d_{\pi',\mu}}[\bar{A}(s)] = \frac{1}{1-\gamma}\sum_{s} d_{\pi',\mu}(s)\,\bar{A}(s)$$

Back to the lemma

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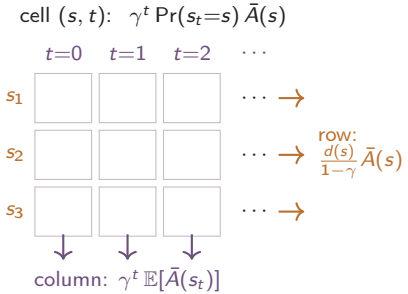
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Sum the grid by rows (step 2)

or by columns (step 3): same total.

Back to the lemma

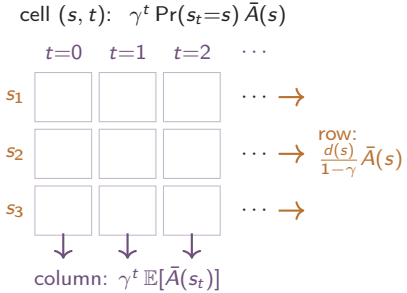
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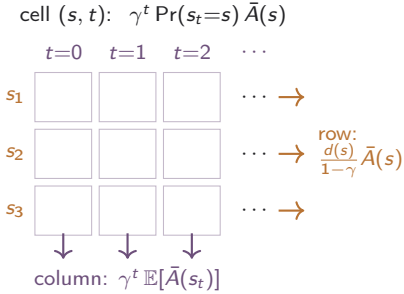
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$\mathbb{E}[\bar{A}(s_t)]$: over $s_t \sim p_t$, $p_t(s) = \Pr(s_t{=}s \mid s_0 \sim \mu, \pi')$ — the state at time t, **not discounted**.
$\mathbb{E}[\sum_t \cdots]$: over trajectories $s_0 \sim \mu$, $a_t \sim \pi'(\cdot|s_t)$, $s_{t+1} \sim P(\cdot|s_t, a_t)$.
The discount sits only in the weights γ^t.

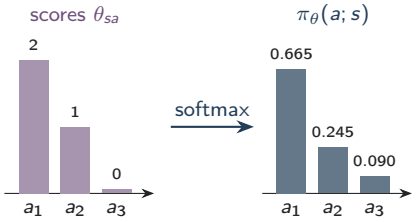


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◂ Back to the lemma

The Gibbs (Softmax) Policy



One state s, three actions. Every action keeps **some** probability.

One parameter per state–action pair ("table lookup"):

$$\pi_\theta(a;s) = \frac{e^{\theta_{sa}}}{\sum_b e^{\theta_{sb}}}$$

- **Why "Gibbs":** the Boltzmann distribution of physics, $p \propto e^{-E/T}$, with $\theta = -E/T$.
- **Always stochastic:** $\pi_\theta(a;s) > 0$; deterministic only as $\theta \to \pm\infty$.
- **Rich:** any policy with full support, in every state independently.
- **Simple score:** $\partial_{\theta_{sb}} \log \pi_\theta(a;s) = 1[a=b] - \pi_\theta(b;s)$

▸ Derivation

◂ Back to policy gradients

Deriving the Gibbs Score

$$\pi(a) = \frac{e^{\theta_a}}{\sum_c e^{\theta_c}}$$

Back to the Gibbs policy

Deriving the Gibbs Score

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Back to the Gibbs policy

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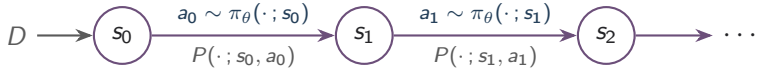
$$\log \pi(a) = \theta_a - \log \sum_c e^{\theta_c}$$

$$\partial_{\theta_b} \log \pi(a) = \underbrace{\partial_{\theta_b} \theta_a}_{1[a=b]} - \underbrace{\frac{e^{\theta_b}}{\sum_c e^{\theta_c}}}_{\pi(b)}$$

Back to the Gibbs policy

Why the Log-Derivative Trick Is Unbiased

θ sits in the **sampling distribution**, not in $R(\tau)$, so averaging $\nabla_\theta R$ gives nothing.



$$p_\theta(\tau) = D(s_0) \prod_t \pi_\theta(a_t; s_t)\, P(s_{t+1}; s_t, a_t)$$

$$\nabla_\theta \eta_D = \sum_\tau \underbrace{\nabla_\theta\, p_\theta(\tau)}_{p_\theta(\tau)\,\nabla_\theta \log p_\theta(\tau)} \qquad R(\tau) = \mathbb{E}_{\tau\sim\pi_\theta}\big[R(\tau)\,\nabla_\theta \log p_\theta(\tau)\big]$$

$$\nabla_\theta \log p_\theta(\tau) = \sum_t \nabla_\theta \log \pi_\theta(a_t; s_t)$$

(log turns $\prod$ into $\sum$; grey terms have no θ)

$\hat{g} = R(\tau) \sum_t \nabla_\theta \log \pi_\theta(a_t; s_t)$, $\tau \sim \pi_\theta \;\Rightarrow\; \mathbb{E}[\hat{g}] = \nabla_\theta \eta_D$ **exactly**, and no need to know P.

Back to PG = SGA

CPI Through the SGD Lens

Take the table $\pi(a; s)$ itself as the parameter: CPI is a **first-order method** on η_μ.

	SGD (non-convex, smooth)	CPI
direction	$\hat{g} \approx \nabla J(w)$	greedy $\pi' = G_\varepsilon(\pi, \mu)$: best linearized direction
step	$w \leftarrow w + a_k\, \hat{g}$	$\pi \leftarrow (1-\alpha)\,\pi + \alpha\,\pi'$ (Frank–Wolfe step)
step size	$\sum_k a_k = \infty,\ \sum_k a_k^2 < \infty$	$\alpha \propto \hat{\mathbb{A}}$: slope over a smoothness bound
noise	averages out over many steps	each step accurate w.h.p. (Hoeffding)
stationary	$\|\nabla J(w_R)\| \le \epsilon$, random iterate	$\mathrm{OPT}(\mathbb{A}_{\pi,\mu}) < 2\varepsilon$, last iterate
cost	$O(\sigma^2/\epsilon^4)$ samples	$72R^2/\varepsilon^2$ updates $\times$ $\tilde{O}(R^2/\varepsilon^2)$ rollouts
global?	$\times$ a stationary point can be bad	✓ Thm. 6.2, up to the mismatch factor

Back to the three questions

CPI Through the SGD Lens

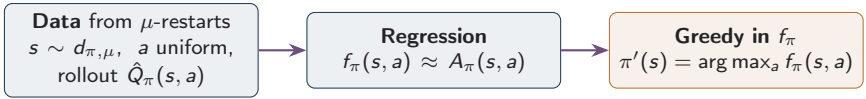
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Same skeleton: a bounded objective and a sure gain per step ⇒ few steps to a stationary point. CPI adds Thm. 6.2: **stationary ⇒ near-optimal**.

Back to the three questions

Building G_ε by Regression on Advantages



Required accuracy (Sec. 7.1):

$$L(f_\pi) = \underbrace{\mathbb{E}_{s\sim d_{\pi,\mu}}}_{\text{average over states}} \underbrace{\max_a}_{\text{worst action}} \left|A_\pi(s,a) - f_\pi(s,a)\right| \le \frac{\varepsilon}{2}$$

Why it suffices: with $e(s) = \max_a |A_\pi - f_\pi|$, $\hat{a} = \arg\max_a f_\pi$, $a^* = \arg\max_a A_\pi$:

$$A_\pi(s,\hat{a}) \ge f_\pi(s,\hat{a}) - e(s) \ge f_\pi(s,a^*) - e(s) \ge A_\pi(s,a^*) - 2e(s)$$

$$\mathbb{E}_{s\sim d_{\pi,\mu}}: \quad \mathbb{A}_{\pi,\mu}(\pi') \ge \mathrm{OPT}(\mathbb{A}_{\pi,\mu}) - 2L \ge \mathrm{OPT}(\mathbb{A}_{\pi,\mu}) - \varepsilon$$

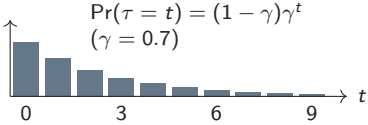
An **average** (ℓ_1) error over $d_{\pi,\mu}$, not the $\max_s$ (ℓ_∞) error of greedy DP (Eq. 3.1): rarely visited states may be fit badly.

Back to CPI

Stopping With Probability $1-\gamma$ Samples $d_{\pi,\mu}$

Run π from $s_0 \sim \mu$; after each step stop w.p. $1-\gamma$, independently of the trajectory. Return s_τ.

$\Pr(\tau = t) = \gamma^t(1-\gamma)$



The discount weights $(1-\gamma)\gamma^t$ are a geometric distribution over t.

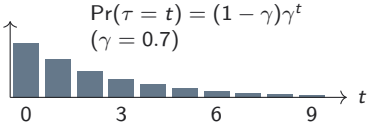
◂ Back to estimating 𝔸

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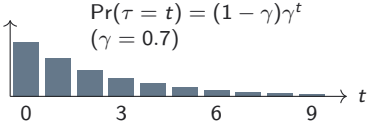
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◂ Back to estimating $\mathbb{A}$

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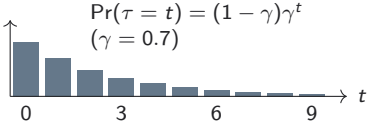
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Step 2 needs independence: τ does not look at the states.
The $(1-\gamma)$ in $d_{\pi,\mu}$ is what makes $\sum_t \Pr(\tau = t) = 1$.
γ = probability of surviving one more step;
$\mathbb{E}[\tau] = \gamma/(1-\gamma)$.



The discount weights $(1-\gamma)\gamma^t$ are a geometric distribution over t.

◂ Back to estimating $\mathbb{A}$

Hoeffding's Inequality and the Estimate of $\mathbb{A}$

If $X_1, \ldots, X_k$ are **independent** and $X_i \in [a, b]$ a.s., then for $\bar{X} = \frac{1}{k} \sum_i X_i$:

$$\Pr\left(|\bar{X} - \mathbb{E}\bar{X}| > \Delta\right) \;\leq\; 2\exp\Big(-\frac{2k\Delta^2}{(b-a)^2}\Big)$$

Back to estimating $\mathbb{A}$

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Conditions for x_i:

- **Independent:** each x_i comes from a fresh μ-restart.
- **Bounded:** $\hat{Q}_i \in [0, R]$, $|\pi' - \pi| \leq 1 \Rightarrow x_i \in [-n_a R,\, n_a R]$, $b - a = 2n_a R$.
- **Right mean:** $\mathbb{E}x_i = \mathbb{A}$ (uniform a, weight n_a); bias of a finite rollout kept below $\varepsilon/6$.

◂ Back to estimating $\mathbb{A}$

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$$\Pr\left(|\hat{\mathbb{A}} - \mathbb{A}| > \Delta\right) \;\le\; 2\exp\Big(-\frac{2k\Delta^2}{4n_a^2R^2}\Big) = 2e^{-k\Delta^2/(2n_a^2R^2)}$$

$$k \;\ge\; \frac{2n_a^2R^2}{\Delta^2}\log\frac{2}{\delta} \quad \text{gives error } \le \Delta \text{ w.p. } \ge 1-\delta$$

Back to estimating $\mathbb{A}$

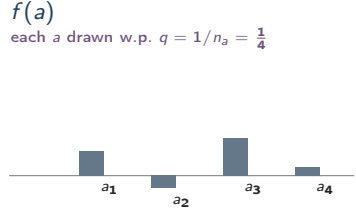
Tool: Importance Sampling

The estimate of $\mathbb{A}$ needs a **sum** over actions, but each rollout tries **one** action.

Sample from q, reweight by $w = p/q$

(needs $q > 0$ where $p > 0$)

$$\mathbb{E}_{a\sim p}[f] = \sum_a q\,\frac{p}{q}\,f = \mathbb{E}_{a\sim q}\big[w\,f\big]$$



We can evaluate p pointwise, but only draw samples from q.

◂ Back to estimating $\mathbb{A}$

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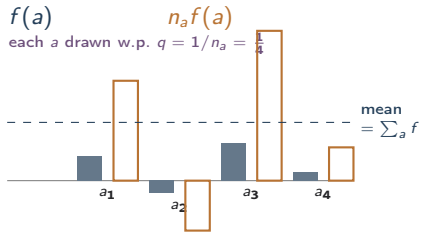
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A plain sum is the case $p \equiv 1$; CPI takes q uniform

$$\sum_a f(a) = \mathbb{E}_{a\sim q}\left[\frac{f(a)}{q(a)}\right] = \mathbb{E}_{a\sim \text{unif}}\big[n_a f(a)\big]$$



Average of the orange bars under uniform a = sum of the blue bars.

◂ Back to estimating $\mathbb{A}$

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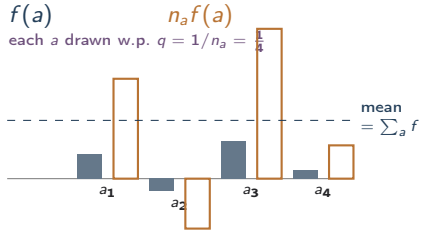
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Unbiased, but the weight stretches the range

$$|f| \le R \;\Rightarrow\; |n_a f| \le n_a R \;\Rightarrow\; n_a^2 \text{ in Hoeffding}$$



One sampled bar can be n_a times taller ⇒ more samples needed. TRPO reuses this with $p = \pi_\theta$, $q = \pi_{\theta_{\text{old}}}$.

◂ Back to estimating $\mathbb{A}$

Continuous States and Actions?

Back to footnote 1: "replace the sums with integrals". **True** — no proof uses finiteness.

Why it carries over

- **Perf. difference**: telescoping, any space.
- **Coupling**: mass $1 - D_{\mathrm{TV}}$ on $\{a = \tilde{a}\}$ — also for densities.
- **Perturbation** (App. B): matrices $\to$ Markov operators, same norms.

What "integrals" hides

1. **Bounded** $\epsilon = \sup_{s,a} |A_\pi(s,a)| < \infty$.*
2. **max** $\to$ **sup** over **all** states, not just visited.
3. **Stochastic** policies: deterministic $\Rightarrow D_{\mathrm{KL}} = \infty$.
4. **Measurable** kernels; ρ_π is a measure.

* Often **vacuous** in continuous control: Gaussian actions with reward $-\|u\|^2$ (their Swimmer) give $\epsilon = \infty$. The practical algorithm drops the bound anyway (mean KL, fixed δ).

Back to Thm 1

One Problem, Several Known Algorithms

Objective	Constraint / penalty	Algorithm
Linear in θ	$\frac{1}{2}\|\theta - \theta_{\text{old}}\|^2 \leq \delta$	Vanilla policy gradient
Linear in θ	Quadratic KL, fixed step $\frac{1}{\lambda}$	Natural policy gradient
Full $L_{\pi_{\text{old}}}(\pi)$	none	Policy iteration
$L_{\theta_{\text{old}}}(\theta)$	$\bar{D}_{\text{KL}} \leq \delta$, enforced	**TRPO**

- NPG vs. TRPO: same direction $A^{-1}g$; TRPO **enforces** the KL constraint each step — matters on larger problems.
- Related: REPS constrains the joint $p(s, a)$; TRPO constrains only $\pi(a \mid s)$.

Back

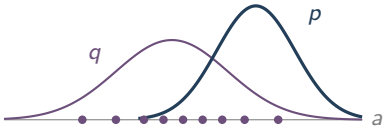
Tool: Importance Sampling

Monte Carlo: an expectation $\approx$ an average of samples

$$\mathbb{E}_{a\sim p}[f] \approx \tfrac{1}{n}\textstyle\sum_i f(a_i), \quad a_i \sim p$$

Samples only from q? Weight $w = p/q$ ($q > 0$ where $p > 0$)

$$\mathbb{E}_{a\sim p}[f] = \sum_a q\,\frac{p}{q}\,f = \mathbb{E}_{a\sim q}\big[w\,f\big]$$



Samples come from q, but we want an average under p.

Back to main

Tool: Importance Sampling

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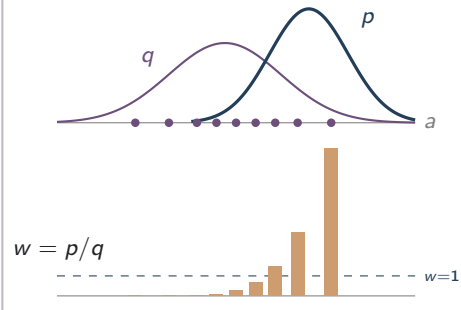
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Unbiased for any q; variance depends on q

$$\hat{\mu} = \tfrac{1}{n}\textstyle\sum_i w(a_i)\,f(a_i), \quad \mathrm{Var}[\hat{\mu}] = \tfrac{1}{n}\mathrm{Var}_q\big[wf\big]$$



p far from q: one rare sample carries a huge weight $\Rightarrow$ high variance.

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Tool: Importance Sampling

Monte Carlo: an expectation $\approx$ an average of samples

$$\mathbb{E}_{a\sim p}[f] \approx \tfrac{1}{n}\textstyle\sum_i f(a_i), \quad a_i \sim p$$

Samples only from q? Weight $w = p/q$ ($q > 0$ where $p > 0$)

$$\mathbb{E}_{a\sim p}[f] = \sum_a q\,\frac{p}{q}\,f = \mathbb{E}_{a\sim q}\big[w\,f\big]$$

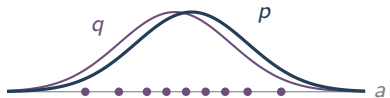
Unbiased for any q; variance depends on q

$$\hat{\mu} = \tfrac{1}{n}\textstyle\sum_i w(a_i)\,f(a_i), \quad \mathrm{Var}[\hat{\mu}] = \tfrac{1}{n}\mathrm{Var}_q\big[wf\big]$$

Policy optimization: estimate expectations under new policy

$$p = \pi_\theta(\cdot \mid s), \quad q = \pi_{\theta_{\text{old}}}(\cdot \mid s)$$

$$w = \frac{\pi_\theta(a \mid s)}{\pi_{\theta_{\text{old}}}(a \mid s)}$$





Trust region keeps $\pi_\theta \approx \pi_{\theta_{\text{old}}} \Rightarrow$ all $w \approx 1$.

Low variance when $q \approx p$.

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Turning Sums Into Samples: The Problem

We have: The local surrogate objective

$$\max_{\theta} \; L_{\theta_{\text{old}}}(\theta) = \sum_{s} \rho_{\theta_{\text{old}}}(s) \sum_{a} \pi_{\theta}(a \mid s)\, A_{\theta_{\text{old}}}(s, a)$$

The problem:

- The sum $\sum_s \rho_{\theta_{\text{old}}}(s)[\cdot]$ requires knowing $\rho_{\theta_{\text{old}}}(s)$ for **all** states
- The sum $\sum_a$ requires enumerating all actions
- For continuous actions, there is no sum at all

What we actually have:

- States sampled from rollouts: $s \sim \rho_{\theta_{\text{old}}}$
- Actions sampled from some policy: $a \sim q$

Next: transform the sums into expectations we can estimate from samples.

Step 1: From State Sum to Expectation

Transform the state sum:

$$\sum_s \rho_{\theta_{\text{old}}}(s)\,[\cdot] = \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim\rho_{\theta_{\text{old}}}}[\cdot]$$

Since $\rho_{\theta_{\text{old}}}$ is a distribution, the sum equals $(1-\gamma)^{-1}$ times the expectation.

Why this helps:

- We can estimate expectations by averaging over samples
- Just run rollouts under $\pi_{\theta_{\text{old}}}$ and collect states
- No need to know $\rho_{\theta_{\text{old}}}(s)$ explicitly

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Why Replacing A with Q is Just a Constant Shift

Substitute $A_{\theta_{old}}(s,a) = Q_{\theta_{old}}(s,a) - V_{\theta_{old}}(s)$:

$$\sum_a \pi_\theta(a \mid s) A_{\theta_{old}}(s,a) = \sum_a \pi_\theta(a \mid s) Q_{\theta_{old}}(s,a) - \sum_a \pi_\theta(a \mid s) V_{\theta_{old}}(s)$$

Since $V_{\theta_{old}}(s)$ doesn't depend on action a, pull it out:

$$= \sum_a \pi_\theta(a \mid s) Q_{\theta_{old}}(s,a) - V_{\theta_{old}}(s) \underbrace{\sum_a \pi_\theta(a \mid s)}_{=1} = \sum_a \pi_\theta(a \mid s) Q_{\theta_{old}}(s,a) - V_{\theta_{old}}(s)$$

The key: $V_{\theta_{old}}(s)$ depends on the **old** policy θ_{old}, **not** on θ we're optimizing!

$$\max_\theta [f(\theta) - \text{constant}] = \max_\theta f(\theta)$$

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